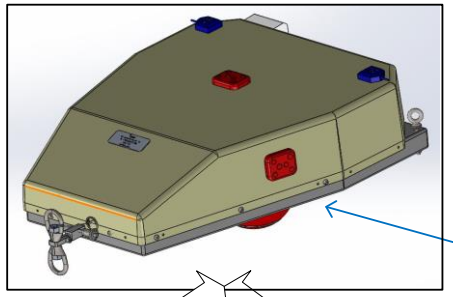
SkyStar 180 - Technical Training

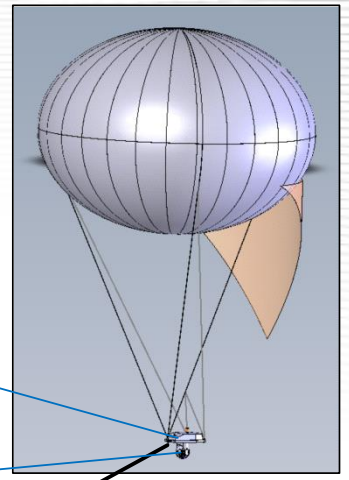
Lesson 1: SkyStar 180 Aerostat Sys. Introduction



SkyStar 180 System review



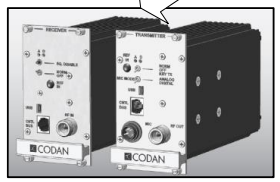
Air Carrier



Aerostat

Ethernet

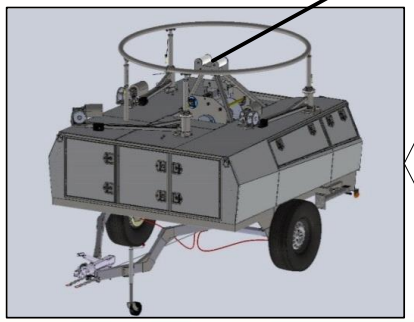
Ethernet





Payloads

Main Tether



Towable Mooring System

Eternet



P.G.C.S

SkyStar 180 System review

Main Parts:

1. 19.5 feet aerostat
2. Airborne carrier
3. EO/IR payload
4. Main tether
5. Mooring system
6. Portable Ground Control Station (PGCS)
7. Electrical system
8. Helium system

SkyStar 180 - Aerostat Training

Lesson 2: SkyStar 180 Aerostat Introduction

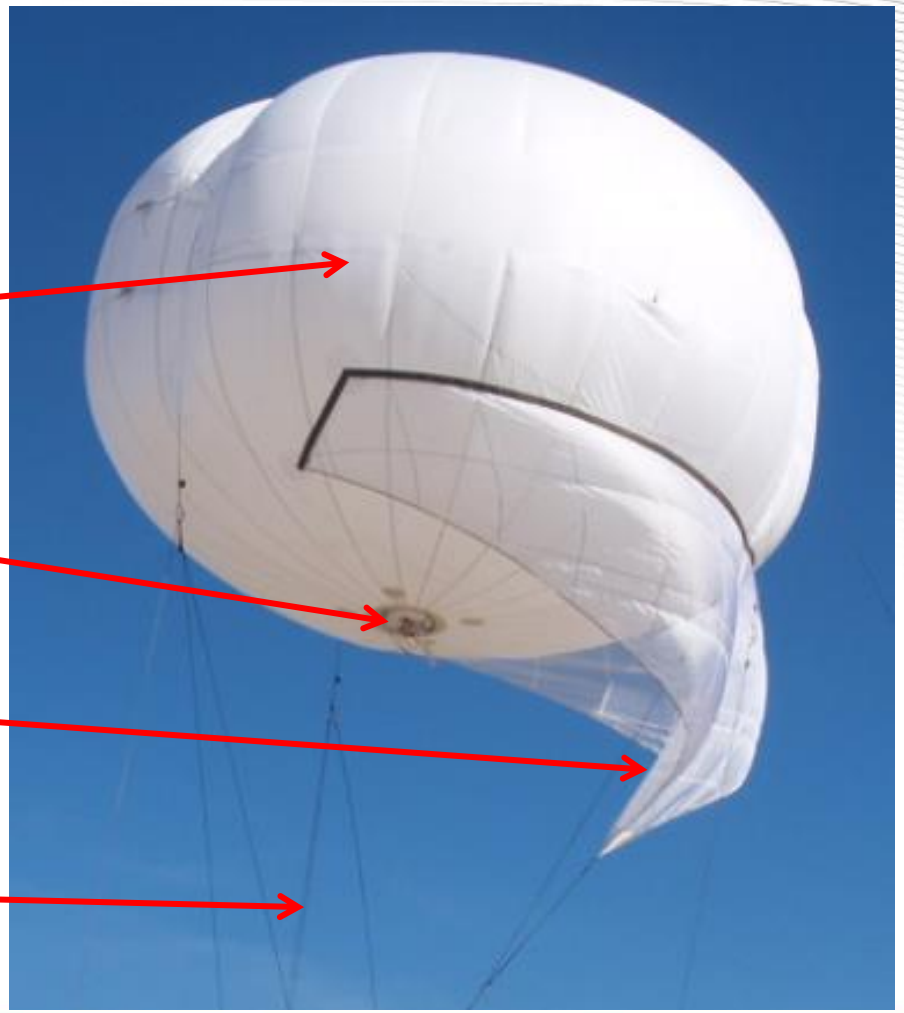
Aerostat Main Parts

Shell

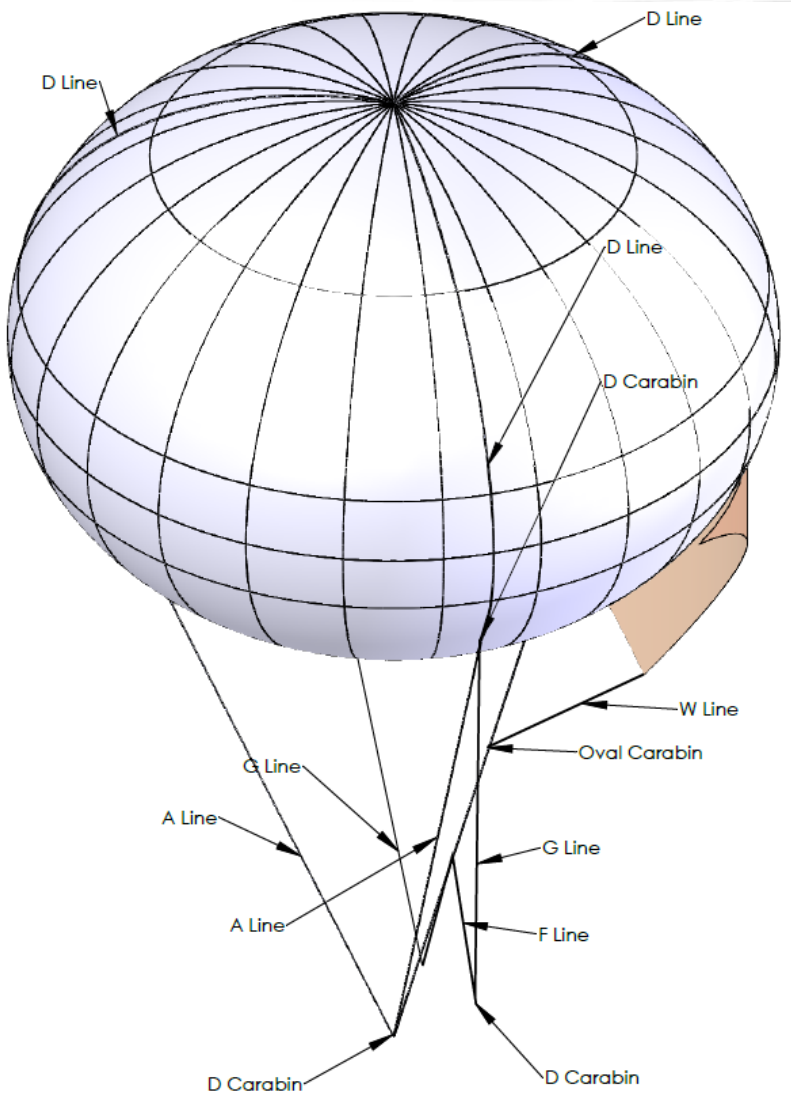
Bladder

Skirt (Wing)

Lines



Aerostat Lines



Aerostat Main Parts

- Shell:
  - Forms the shape of the aerostat and enclosed the bladder inside.
  - The outer fabric layer protects the inside bladder from damage and environmental conditions and provides mechanical strength to the aerostat.
- Bladder:
  - Provides maximum flexibility and expansion.
  - Filled with Helium.
  - The Bladder's material is very resistant to Helium leakage.
- Skirt:
  - Used to stabilize the Aerostat against the wind.
  - Contributes to adding dynamic lift generated by wind flow.
- Lines:
  - Are Used to hold and the aerostat and determine the pitch & roll angles during flight.
  - Used to connect the air carrier and moor it in the ground.

SkyStar 180 – Technical Training

Lesson 3: SkyStar 180 Air Carrier

SkyStar 180 – Air Carrier Main Parts

- **Payloads** –
  - Electro-optic observation system that combines Day, Night and Laser Range Finder sensors.
  - Radio Repeater.
- **Air carrier body -** Used for mechanical and electrical connection of the payloads.

  The carrier includes: Chassis, hood, Power supplies, Communication, Tether's connector, Swivel and D-shape carabiners.
- **Power -** Power supplies provides the electronic devices operating voltage.
- **Video and communication** - Enables the "dialogue" between the payload and the Portable Ground Control Station(PGCS)
- **Radio System** - The customer's equipment, used for communication, is installed together with the electro-optical payload.

Air Carrier – Epsilon 140 Payload

- Three-sensor Stabilized Miniature Payload
  - Day TV-CCD, HD 720p 1270x720px, Continuous Optical Zoom x30, FOV: 1.3°-37.9°.
  - FLIR-CCIR, 3-5 micron, SD 640x512px, Continuous Optical Zoom x3.3, FOV: 7.7° - 24.5°.
- Operating Voltage 22-26VDC.
- Video & Communication protocol: Ethernet.
- Weight : 1.7 Kg.
- Day Detectability Man 5Km , Nato 15Km.
- Night Detectability Man 2Km Nato 2.7Km.



Air Carrier – Power Supplies

- 220VAC via the tether through the power connector:
  - E.O.P (Ethernet Over Power) device.
  - AC to DC converters.

- AC to DC converters:
  - 220VAC to 24VDC conversion to power the E/O payload ,strobe lights and fan.
  - 220VAC to 12VDC conversion to power Radio Repeater.

Air Carrier – Video, Communication & Peripheral Components

Payload - Receiving commands from the PGCS and execute the command by the relevant operation: left, Right, Day, Night camera, Zoom And so on.

The payload also has a navigation system which is a major factor in the reliability of the carrier's spatial orientation, and target acquisition.

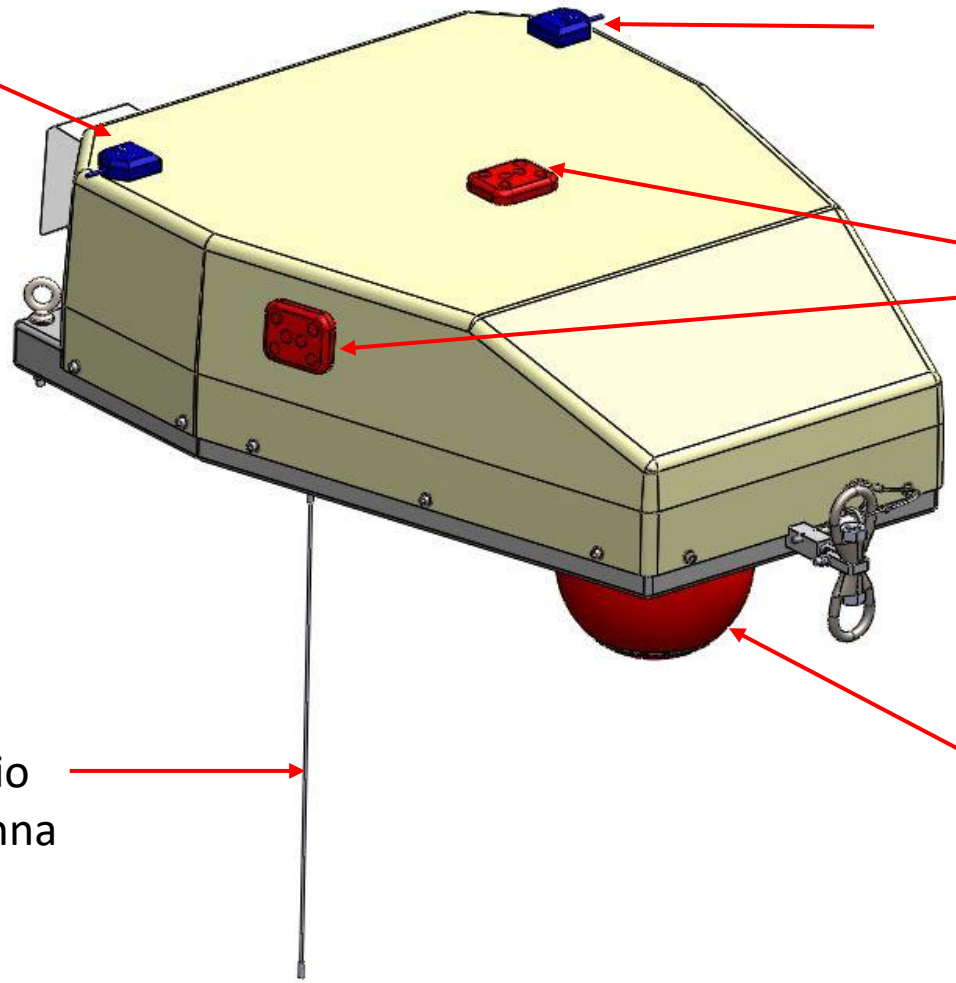
The payload Command and video outputs are through Ethernet line.

E.O.P device – modulate Ethernet communication (from the payload) on the power lines.

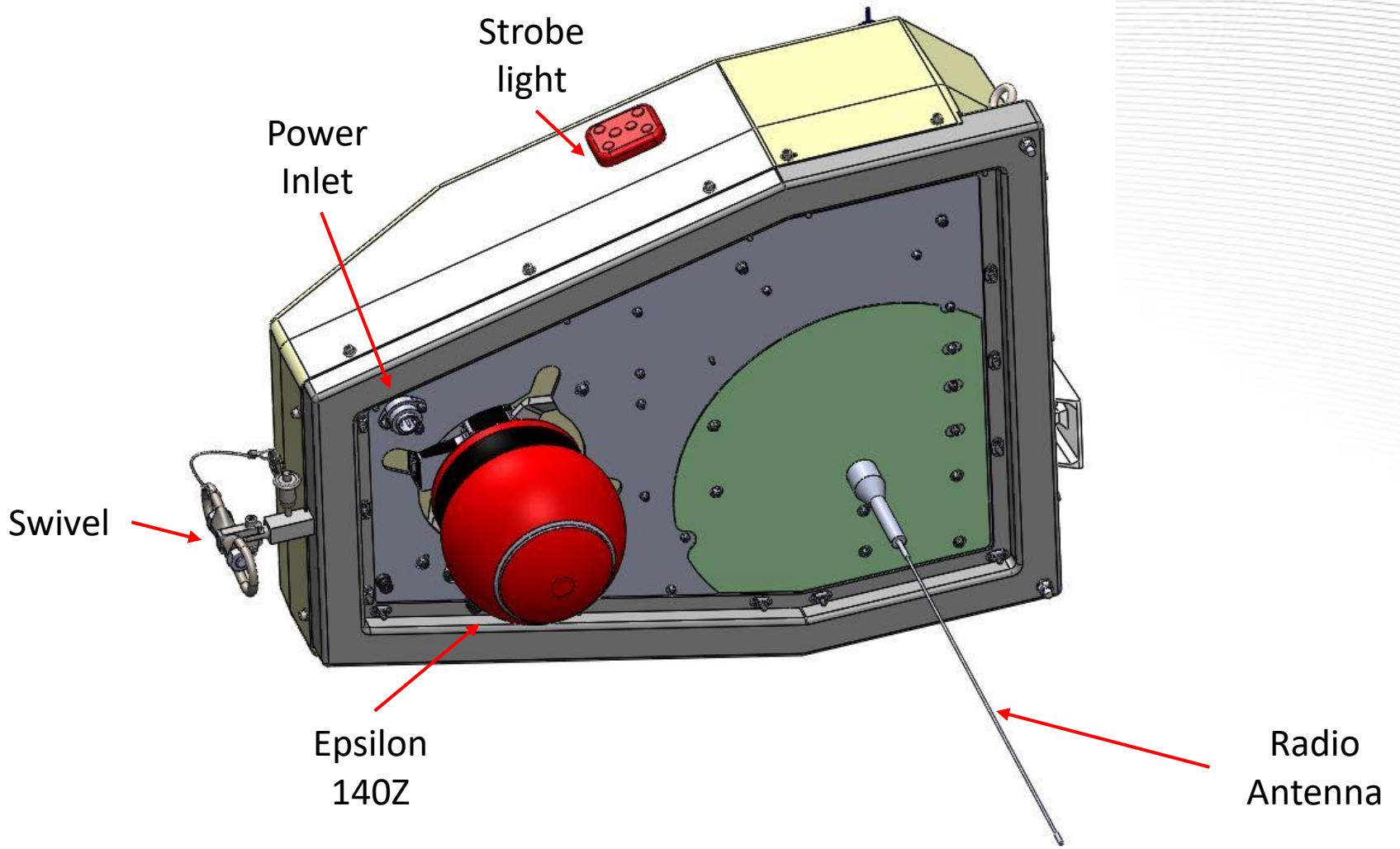
Strobe lights - Increases the aerostat visibility required according the regulation of air traffic.

Swivel – Mechanical connection between the Aerostat Lines, the tether & Air carrier.

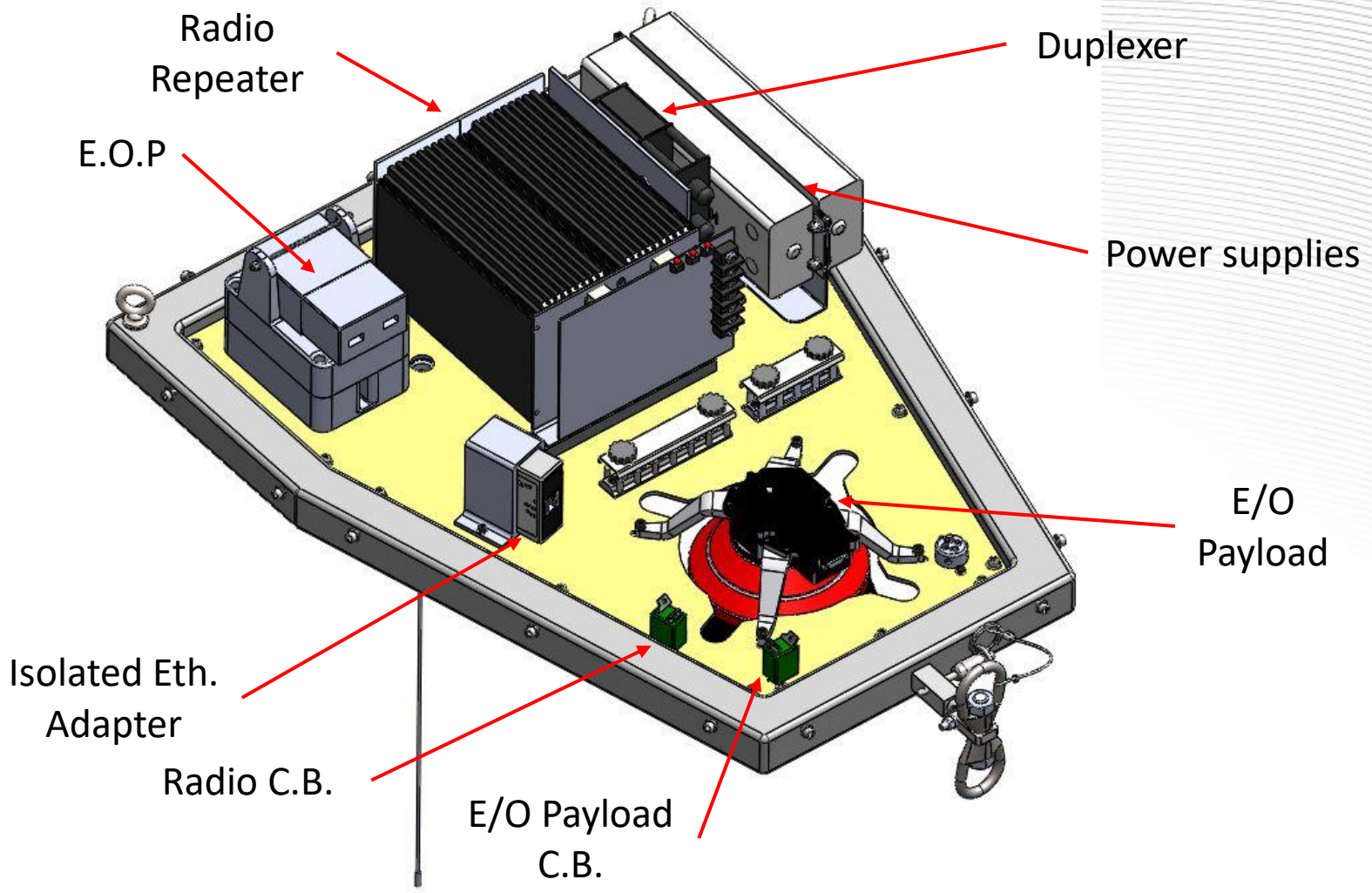
Air Carrier – Top External Components



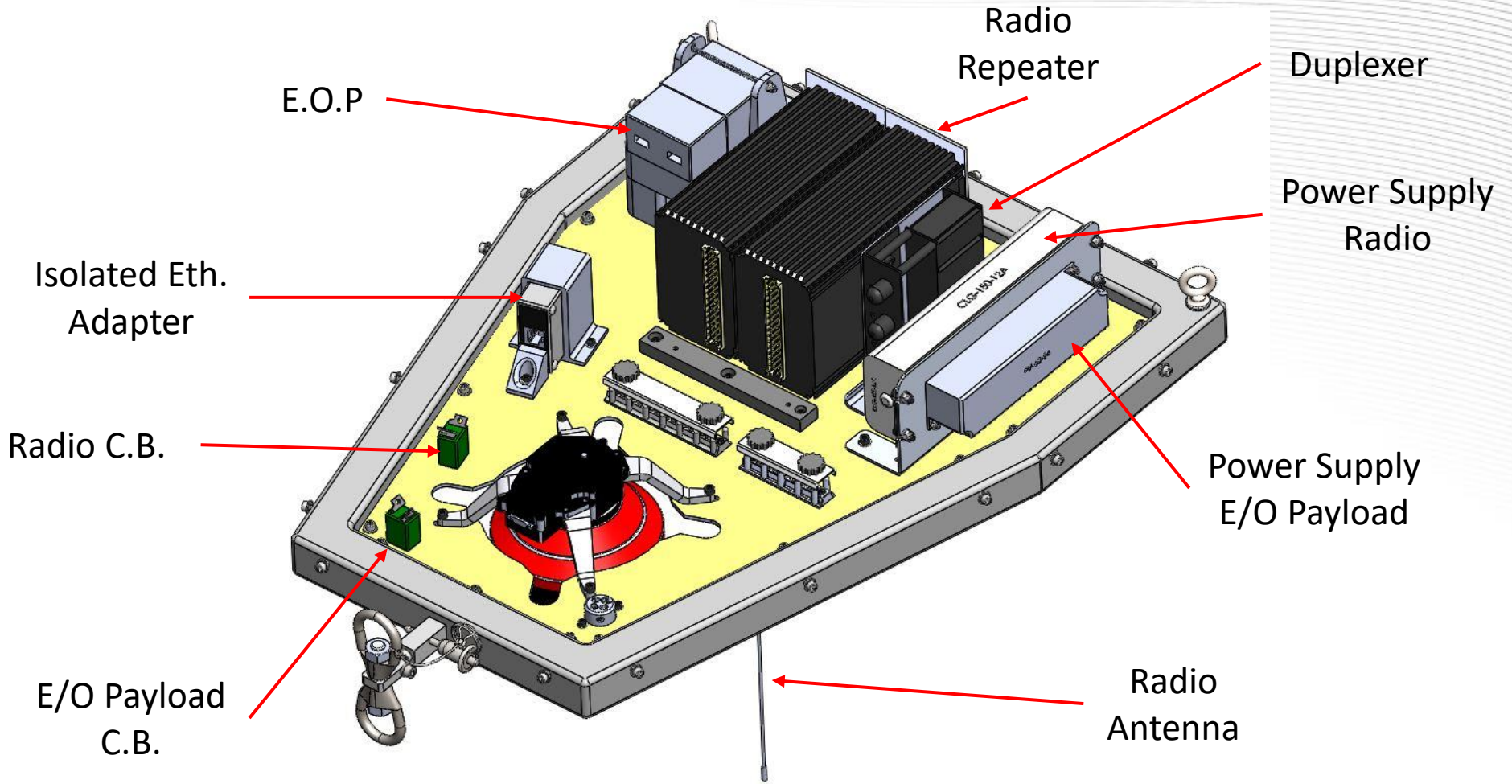
Air Carrier – Bottom View



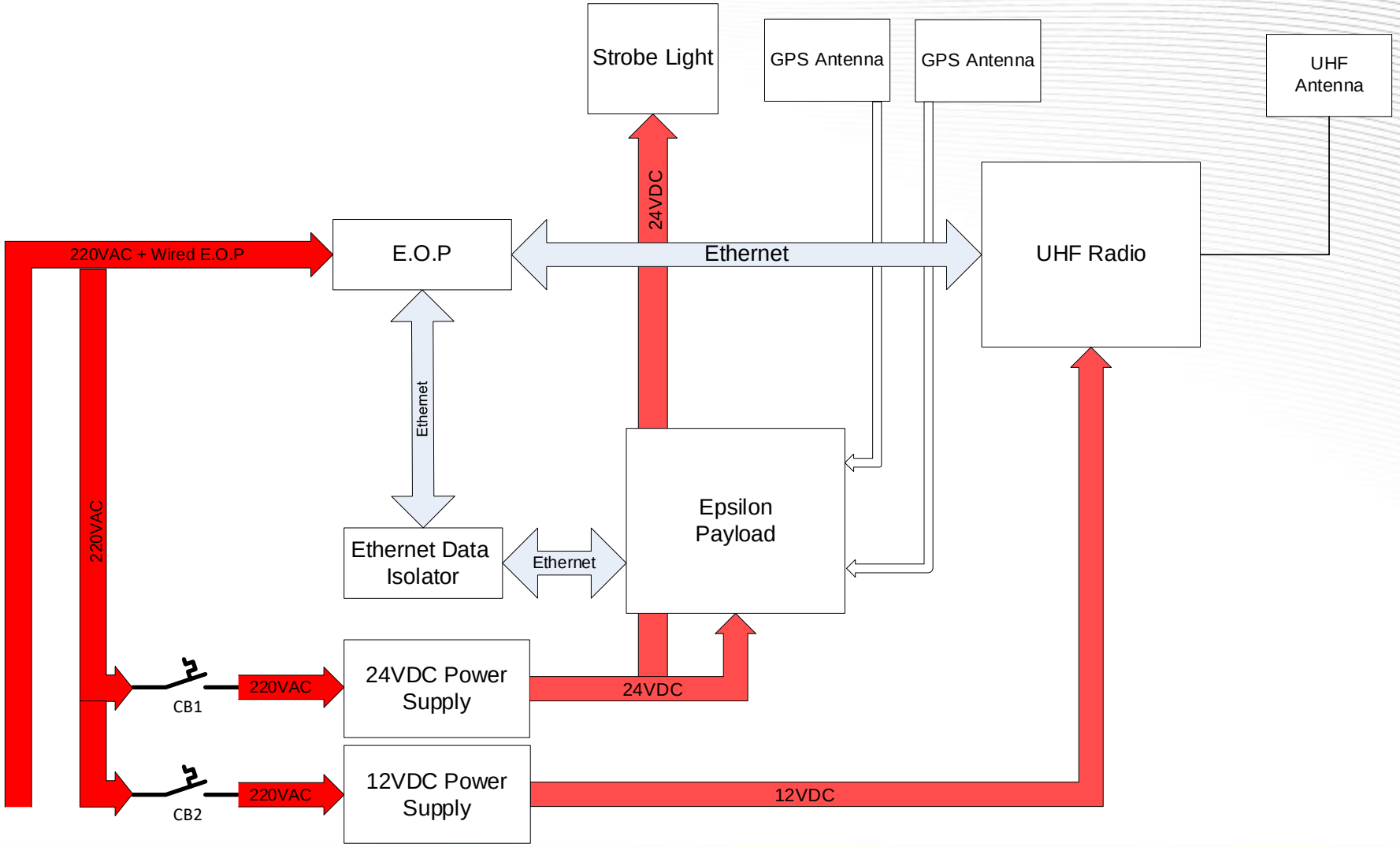
Air Carrier – Payload & Parts Assembly



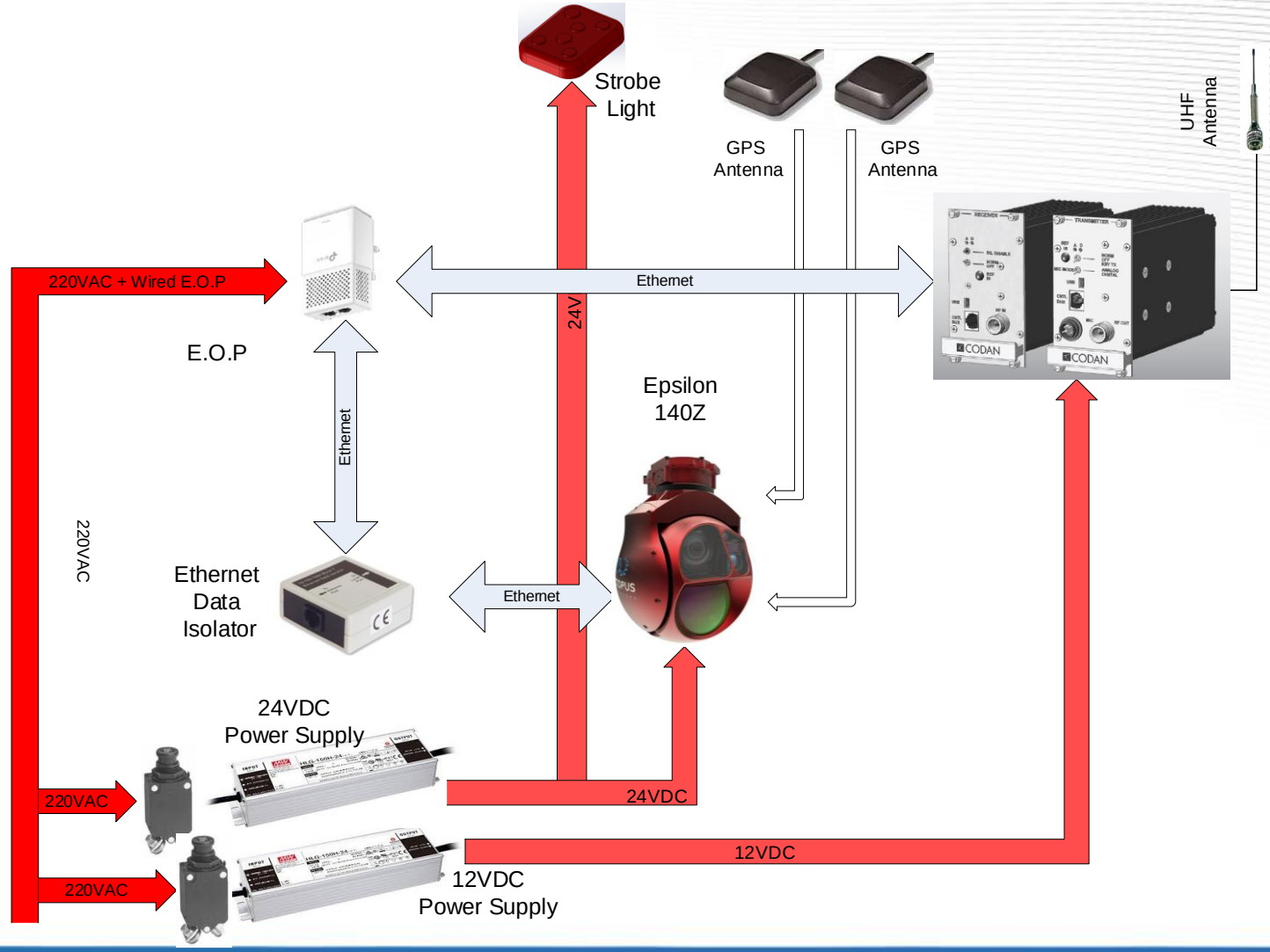
Air Carrier – Payload & Parts Assembly



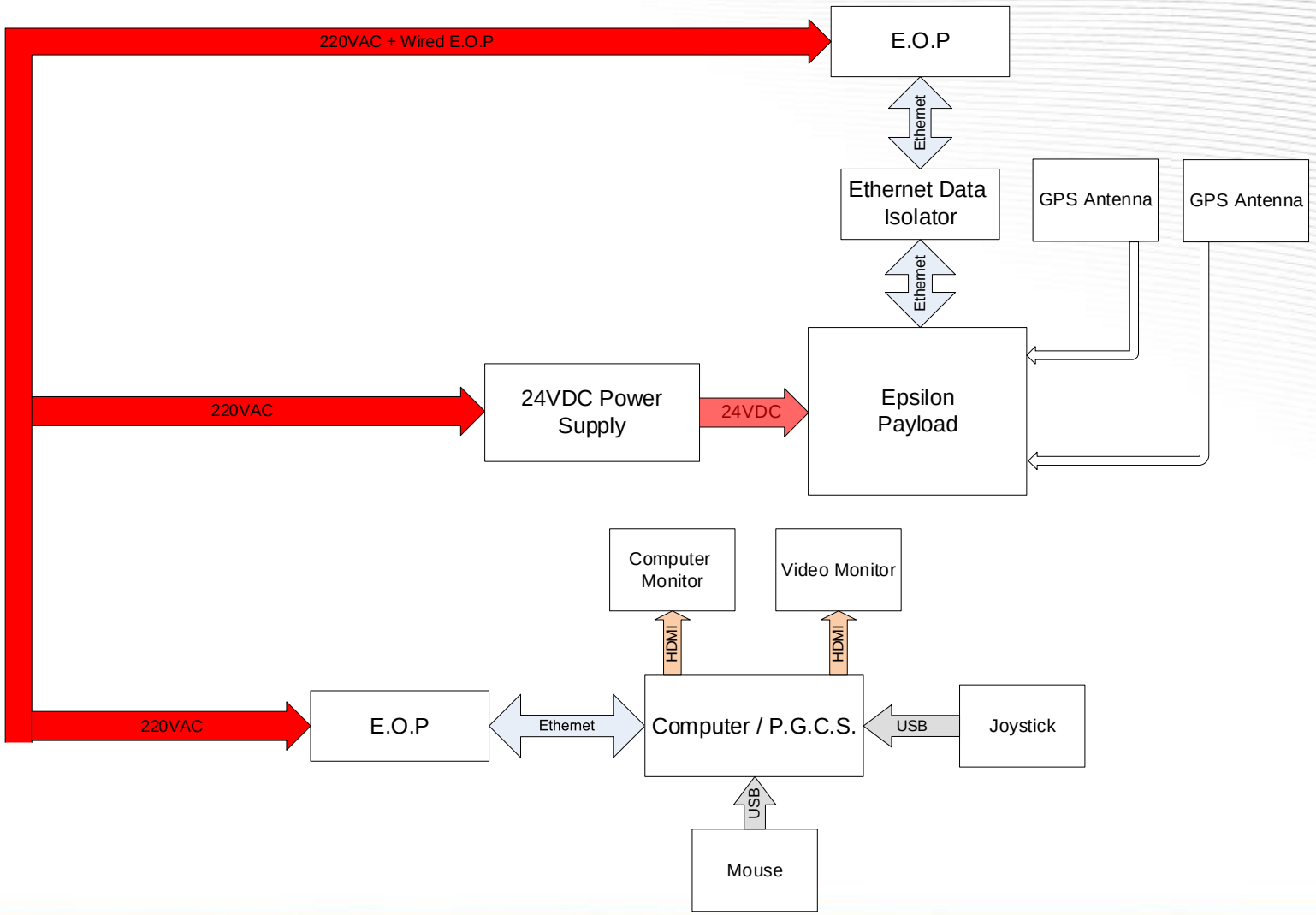
Air Carrier – Block Diagram



Air Carrier – Illustration Diagram



Epsilon Payload ATP Connection



SkyStar 180 – Technical Training

Epsilon Air carrier

SkyStar 180 – Technical Training

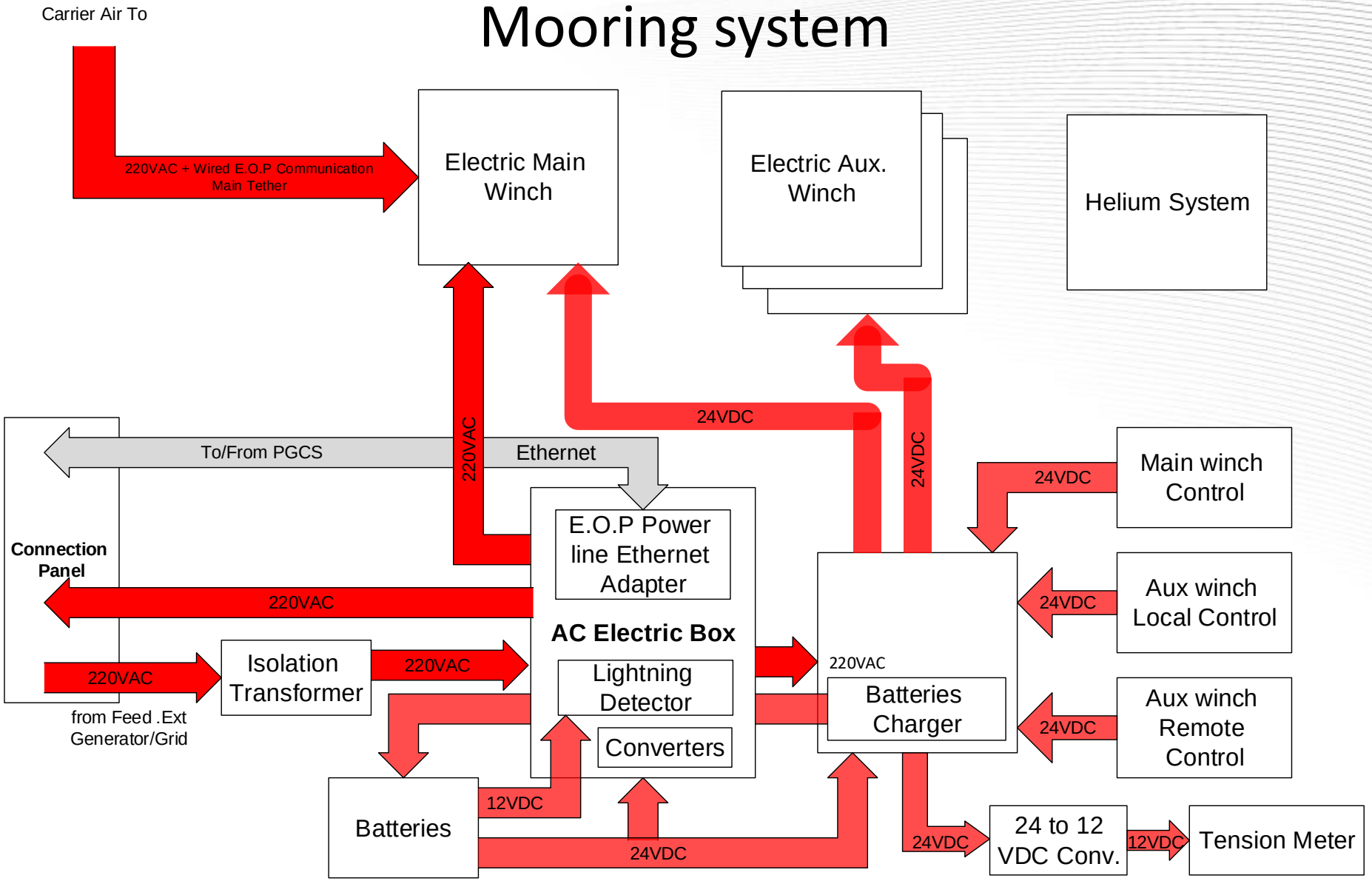
Lesson 4: SkyStar 180 Mooring System



Mooring system

- The mooring system is a ground unit which anchors the aerostat while it is hovering or while the aerostat is moored.
- The mooring system comprises of supporting equipment that allows inflation, launching and recovering of the aerostat.
- The mooring system is also comprises the electrical and helium systems equipment, arranged in dedicated cabinets.
- The mooring system comprises of the following units:
  - Towing Frame (Trailer).
  - Mooring ring.
  - Main winch.
  - 3 auxiliary winches.
  - Helium Manifold for 6 Cylinders.
  - Electrical compartment.
  - Tether Tension Meter.

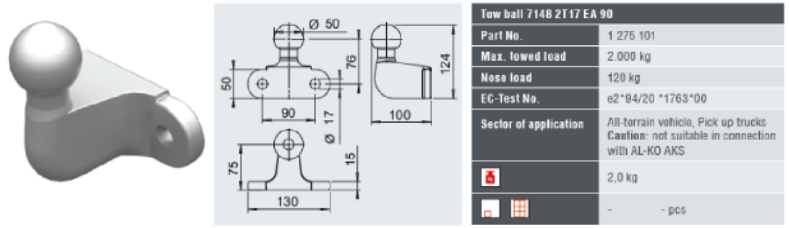
Mooring system



Mooring system parts

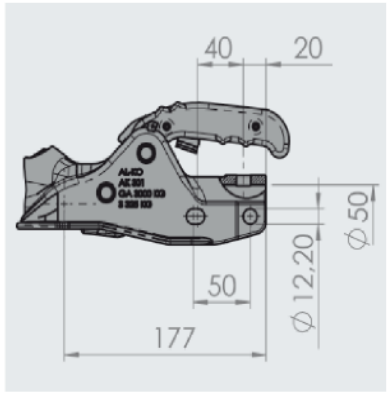
Towing Frame (Trailer):

Total Wight: 1500Kg



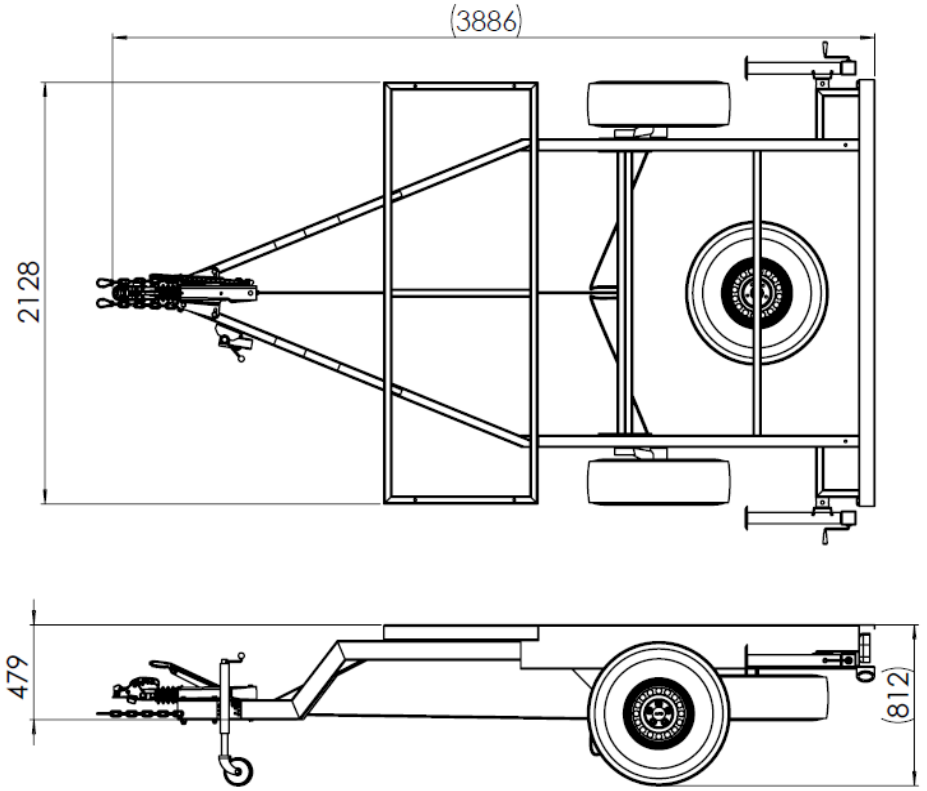
Tow ball 7148 2T17 EA 90	
Part No.	1 275 101
Max. towed load	2.000 kg
Nose load	120 kg
EC-Test No.	e2*94/20 *1763*00
Sector of application	All-terrain vehicle, Pick up trucks **Caution:** not suitable in connection with AL-KO AKS
	2,0 kg
	- - pcs

Vehicle Towing Coupling - Tow Ball ISO DIA. 50mm



Ball-type standard coupling Profi V AK 301	
Part No.	1 221 746
Max. total weight	3.000 kg
Nose load	325 kg
Fixing holes	horizontal + cross
Material	Cast Iron
EC-Test No.	e1 00-1594
Torque maximum	86 Nm
	2,85 kg
	200 pcs

Towing Coupling on Trailer - AK 301 or Compatible



Mooring system parts

Mooring Ring:

Ring diameter: 1900 mm.

Diameter of rod: 1.5“ Soft and protective cover.

4 telescopic poles:

Minimum height: About 853 mm.

Maximum height: About 1700 mm.

Method for determining height: Pins and holes.



Mooring system parts

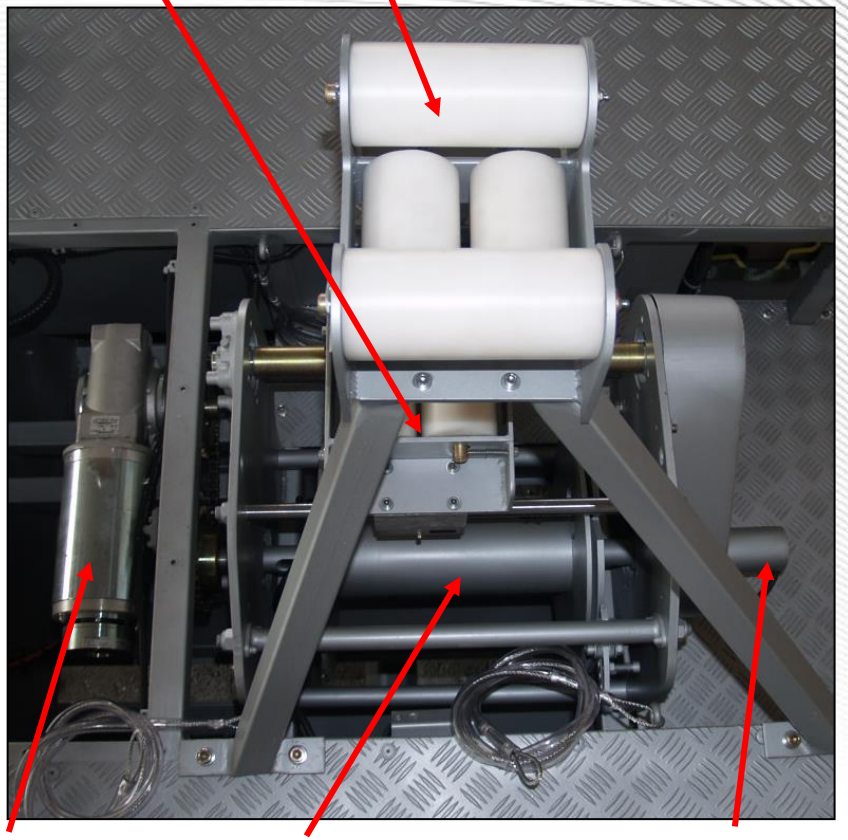
Main Winch:

- The main winch has a level wind, uses to wind the main tether upon it. The main winch is used to control the releasing and recovering of the main tether.
- The main winch is controlled by Operation Handle, located at the DC control panel.
- The main winch comprises of: Electrical Motor, Storage Drum, Guide Rollers mechanism (comprises of a guide rollers and a level wind) and electrical slip-ring.



Operation Handle

Safety Switch



Guiding Rollers

Motor

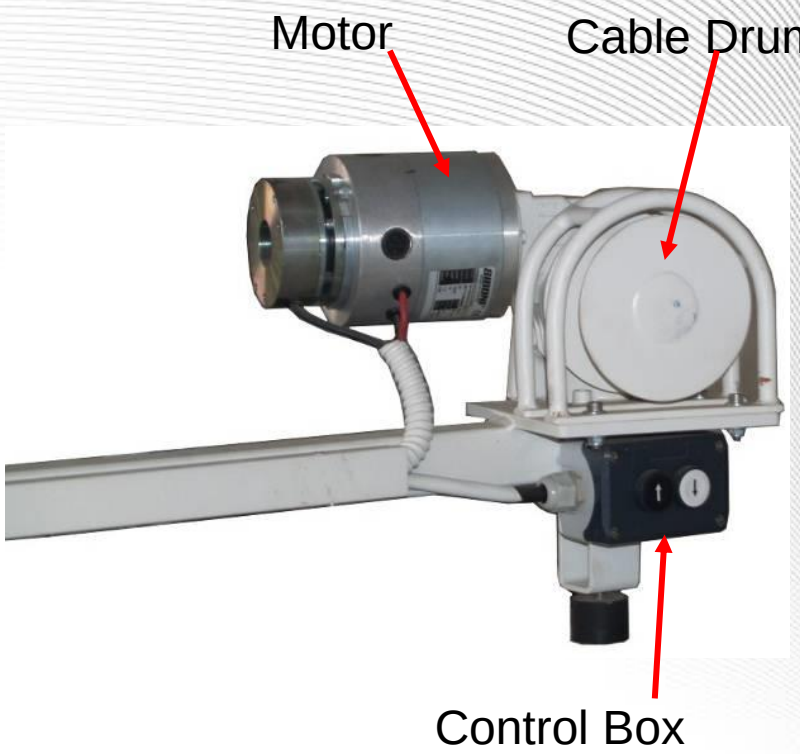
Storage Drum

Slip Rings

Mooring system parts

Auxiliary winches:

- Three auxiliary winches assist handling the aerostat during launching and recovering processes, using three close-hold (CH) lines. Each winch comprises of: Electrical Motor, Cable Drum and local control box.
- Local Control Box uses for controlling the auxiliary winch operation: UP – turning the winch in order to free the Close-Hold (CH) line, DOWN – turning the winch in order to pull the CH line and wind it onto the Drum.
- A remote control used to operate the 3 Aux. winches together from a common place.



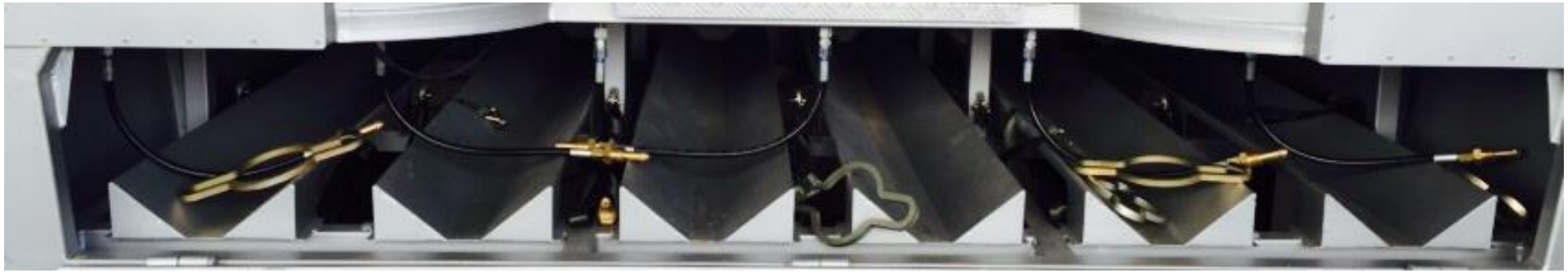
Working forces:

- Operation load: up to 150 kg weight
- Working speed: 12 m/min.



SkyStar 180 - Aerostat Training

Lesson 5: SkyStar 180 Helium Sys.

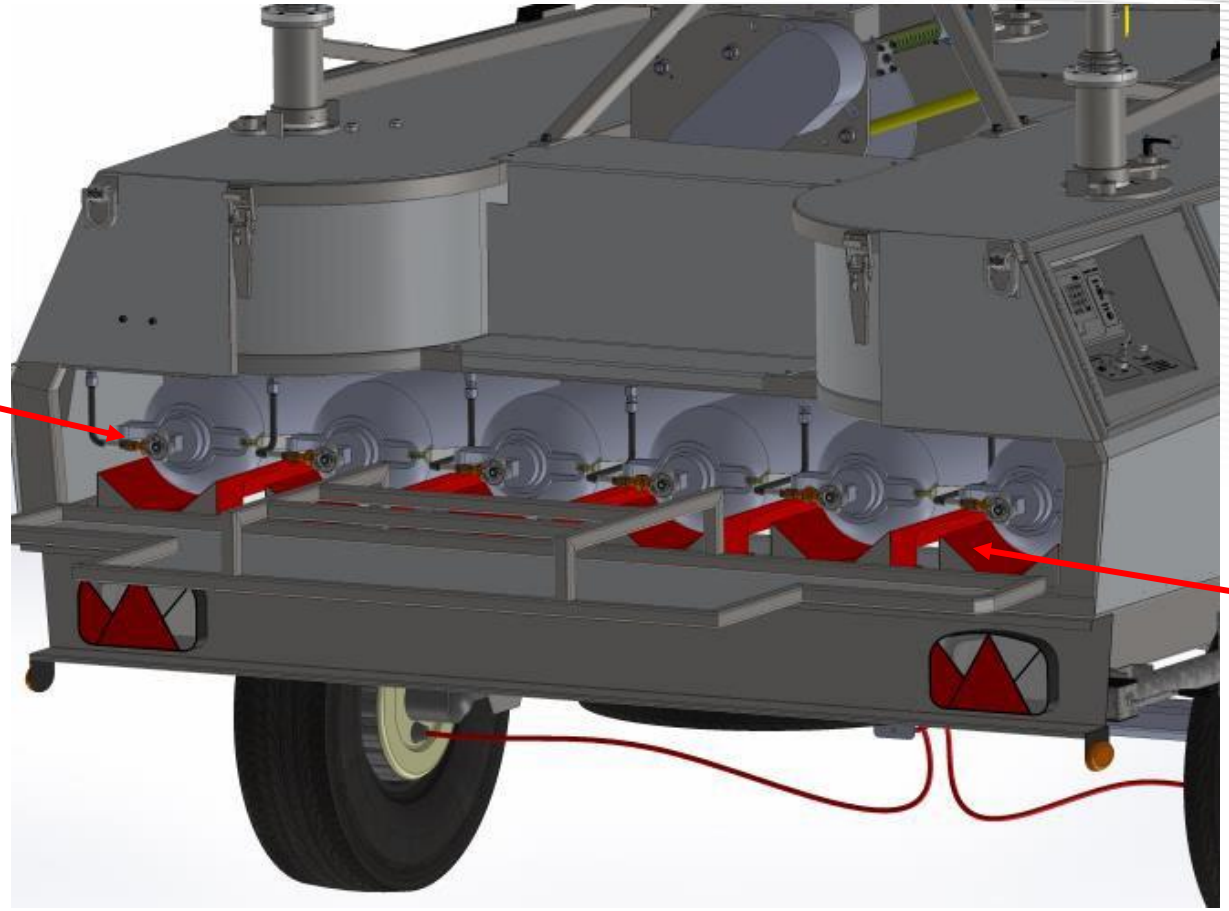


Mooring system – Helium parts

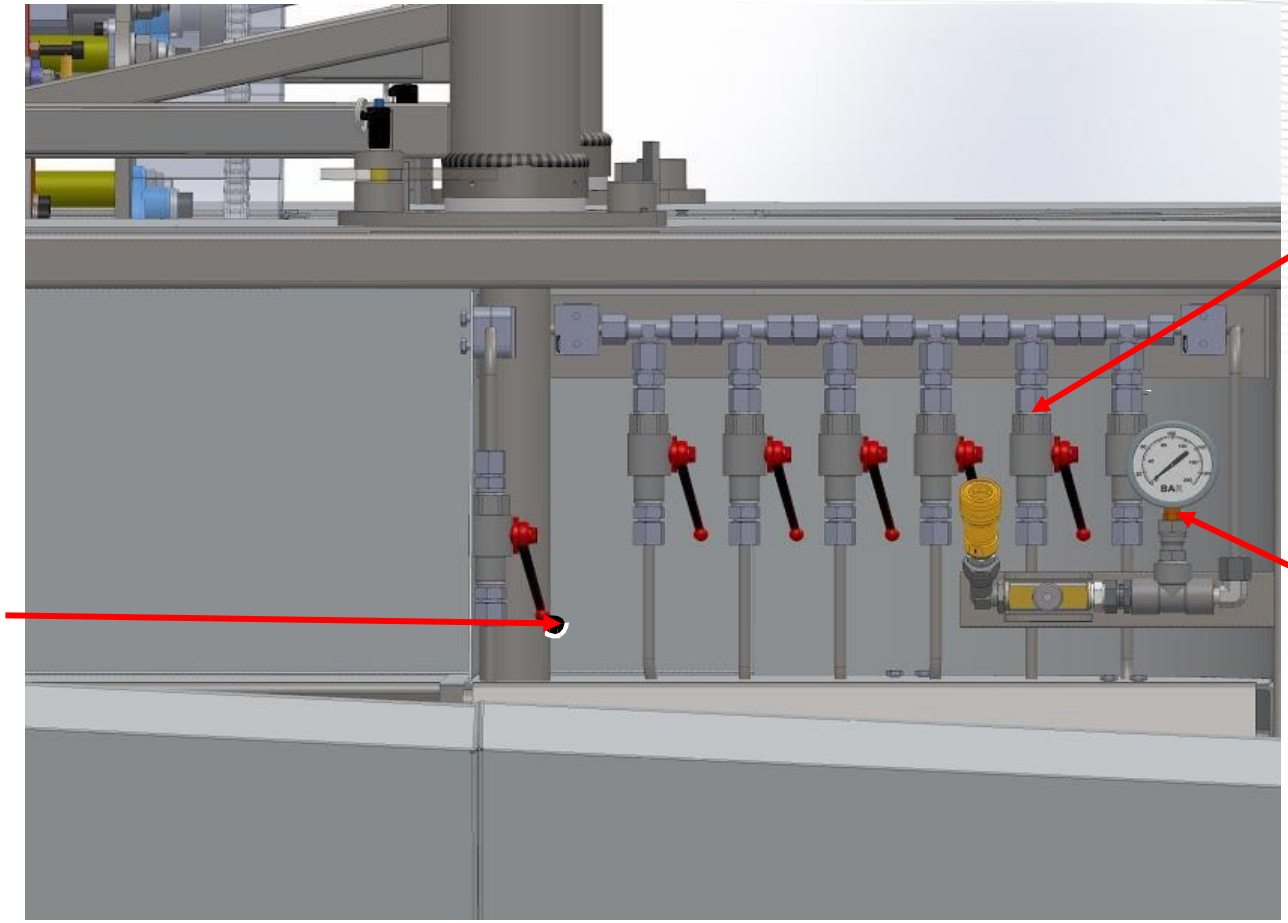
Helium Manifold for 6 Cylinders:

- Used for aerostat inflation and Helium filling.
- Comprises of tubes, connectors, valves and Helium Cylinders, designed to work under high pressure.
- The Helium system provides:
- Safe Storage – Helium has the advantage of being non-flammable and non-toxic. Helium gas is stored in pressurized cylinders carried inside the frame assembly.
- Easy inflation – The system utilizes a quick connection coupling (valve) connected to the aerostat during inflation.
- Pressure Indication – the Helium pressure in the system is indicated by a pressure gauge, located near the main valve.
- Helium flow control – Helium flow is controlled by means of cylinders primary valves, secondary valves, pressure discharge valve and main valve.

Mooring system – Helium parts



Mooring system – Helium parts



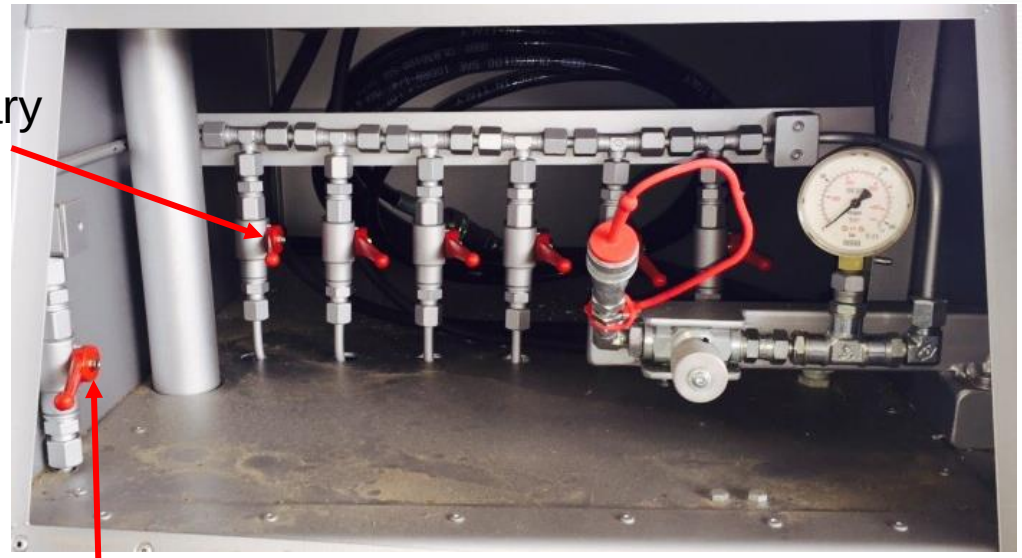
Secondary Valve

Pressure Relief Valve

Pressure gauge

Mooring system - Helium parts

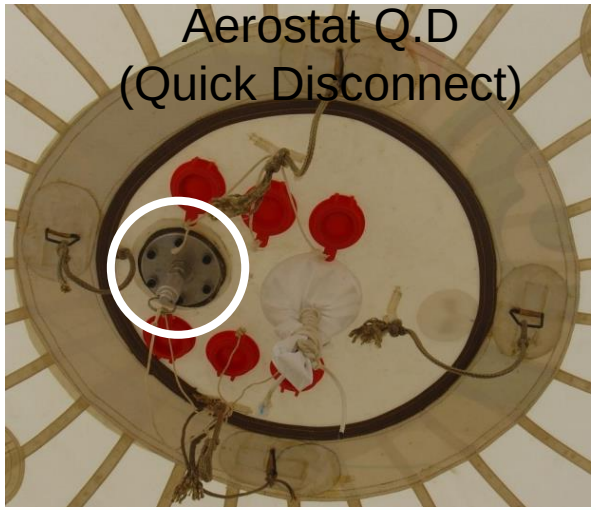
Secondary Valve



Relief Valve

Cylinder Valve



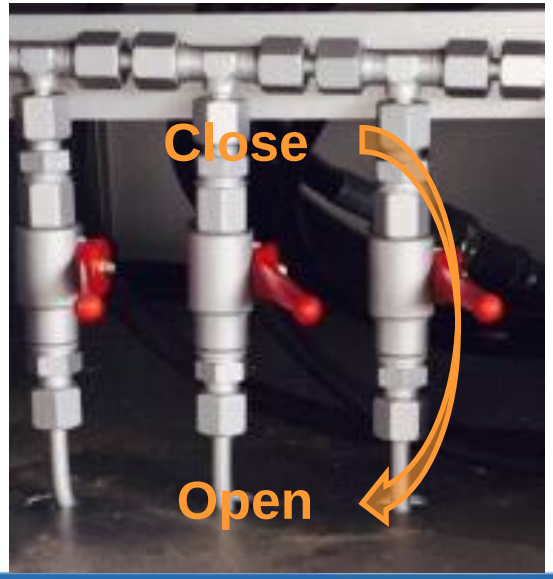




Mooring system - Helium parts

Helium Pipe Lines:

- On the mooring station there are flexible and rigid pipes.
- Every Cylinder is connected by a flexible pipe through a Secondary Valve to a rigid Manifold.
- The rigid pipeline leads to the aerostat helium filling Q.D. (Quick Disconnect) connector.
- The system and its parts were designed and tested for operating in pressure of 200 bar.



Mooring system - Helium parts

Helium Fueling Hose:

- Connects the manifold of the Mooring Sys. to the aerostat.
- Both ends are terminated with Q.D (quick disconnect) connectors.



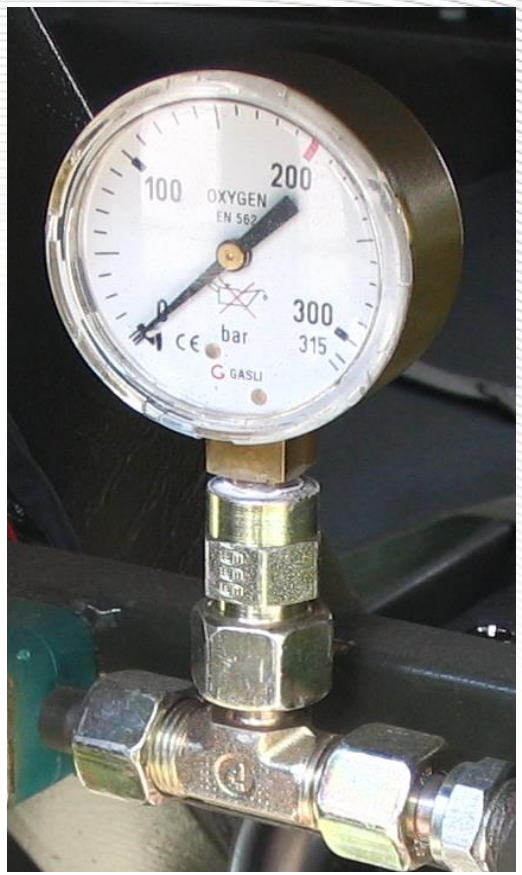
Female Q.D (quick disconnect) connector



Mooring system - Helium parts

Pressure gauge:

- Measures the pressure in the Helium Manifold and Cylinders.
- While measuring, the main valve must be closed.



SkyStar 180 - Aerostat Training

Lesson 6: Main Tether

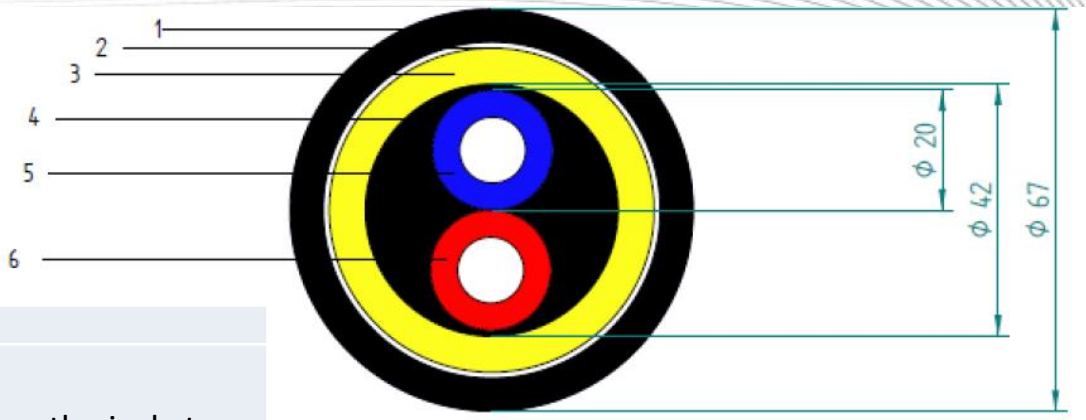
Aerostat Main Tether

The Main Tether is the main cable that mechanically holds the aerostat and connects the Air Carrier, to the Mooring System. The Main Tether carries the power and data communication: receiving payload uplink commands and transmitting downlink video images and data between the airborne carrier, via the mooring system, to Portable Ground Control Station (PGCS).

The Main Tether is connected to three different parts:

1. Main winch – the lower side of the Main Tether is (mechanically) connected to winch's drum and (electrically-through slip rings/rotary joints) to the AC electric box.
2. Aerostat's Swivel – Main Tether is secured mechanically to the lower eye of the aerostat's swivel (the second connected to the Aerostat’s flying lines).
3. Airborne carrier – The Main Tether electrical connector is connected to the Airborne Carrier connector.

Aerostat Main Tether – General Structure



#	Part	Function
1	Outer PU Jacket	Overall black UV and ESD resistant PU compound outer jacket. Protects from mechanical damages and weather conditions. In addition the jacket has a carbon additive which enables the discharge of electrical charges and static electricity to the ground.
2	Separation Tape	A plastic material (Mylar) prevents the jacket from attaching to the Kevlar fibers.
3	Aramids Reinforce	A mash (braid) of Aramids around the electrical core. Strong fire-resistant synthetic fiber derived from polymer chain, interweaved with one another which encircle the core and form a symmetrical distribution of the forces around the cable.
4	Inner PU Jacket	Same as outer PU jacket (1).
5,6	Copper	One twisted pair of 20 AWG strand low capacitance conductors for data and power transmission, colored red x blue. Carry the 230V AC electrical power inside the cable.

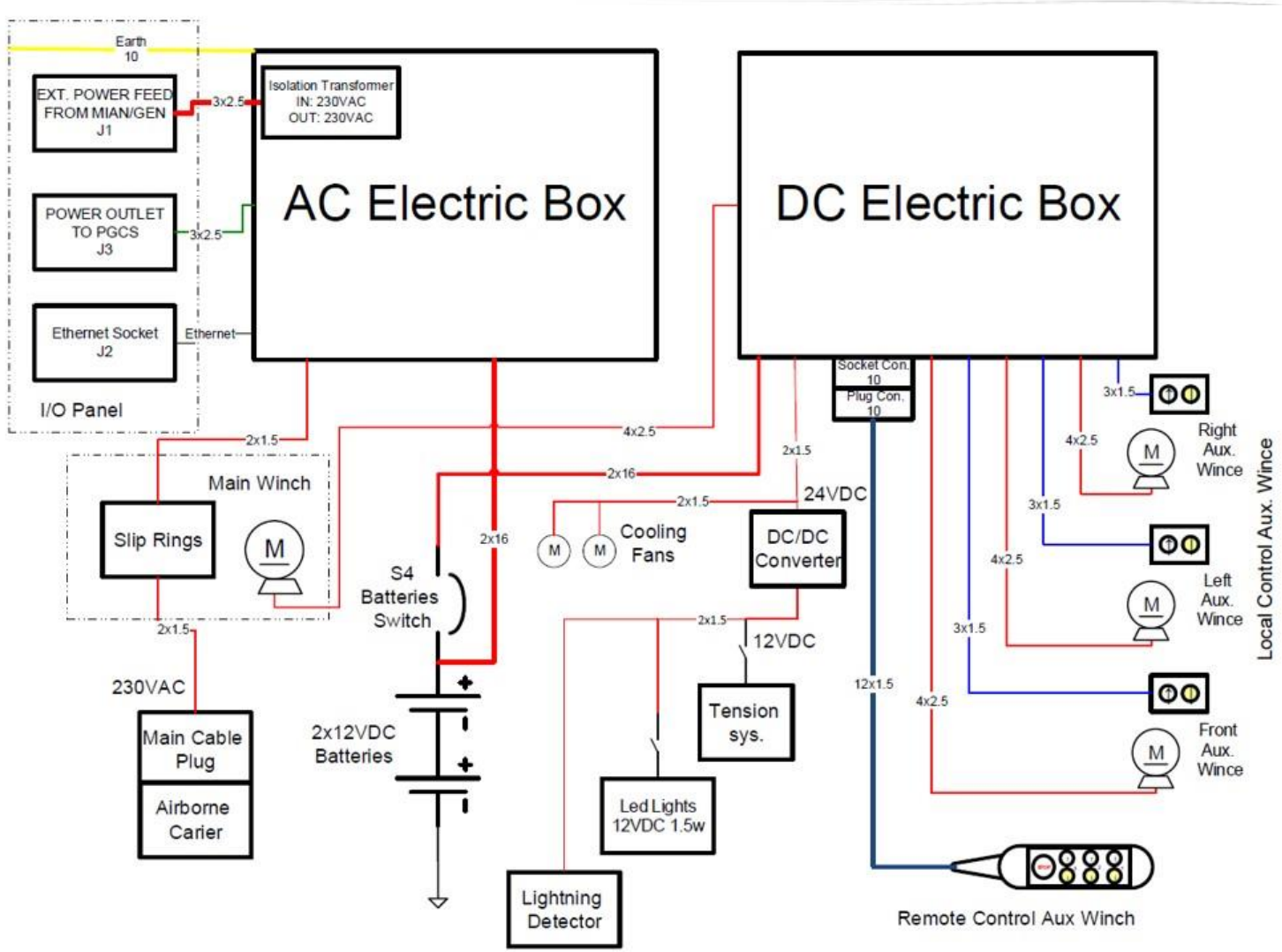
SkyStar 180 - Aerostat Training

Lesson 7: Electrical System

Mooring system – Electric Sys. Description

- 2 batteries are used as main power source to feed the system.
- The system has input (single phase) from external source to charge the batteries and to use as an alternative power source
- The external power source connected through an isolation transformer.
- With this isolation transformer provides floating power. (galvanic isolation between Neutral and Ground-IT system).
- High level of operational and electrical system safety in ungrounded power supply systems achieved by using an Isometer which continuously monitors the insulation resistance between system and earth. The Isometer monitor produces audio-visual warning and indication.
- A bi-polar main breaker (16 Amps) protect and disconnect the external power source.
- All the systems consumer have bi - polar circuit breaker (character Z) to protection and disconnect, with lower current level (to enable selectivity).
- 230VAC produced from the batteries with two alternative DC/AC Inverters (One active and one backup)

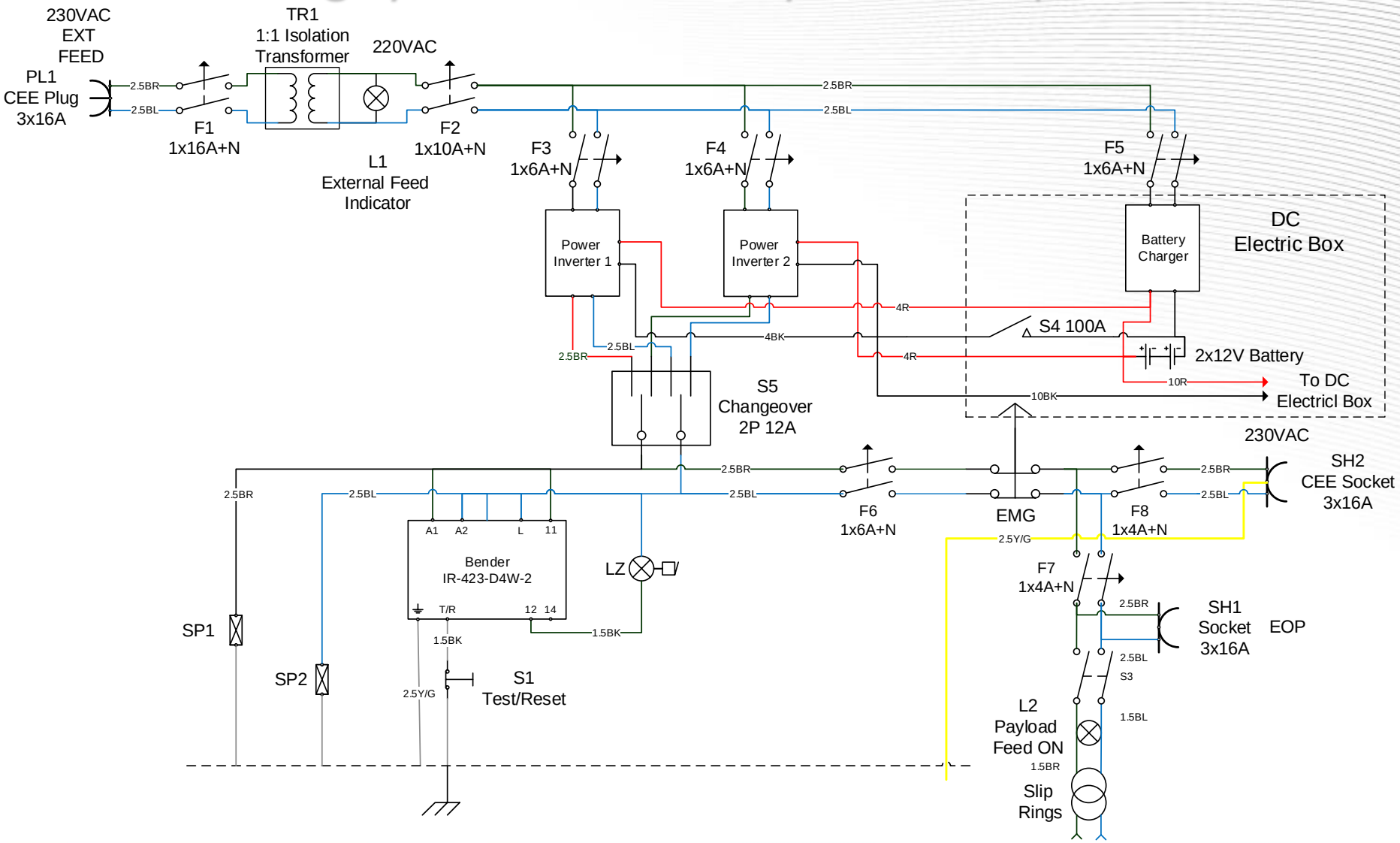
Mooring system – Electric Sys. Block Diagram



Mooring system – Electric Sys. AC Compartment



Mooring system – Electric Sys. AC Compartment



Mooring system – Electric Sys. AC Compartment

AC Cabinet



External feed(F1)

Isolation Transformer output (F2)

Surge protection

Isometry (isolation tester)

Inverters out circuits breaker (F6)

Inverters AC input circuits breaker (F3,F4)

Power output Inverters changeover (S5)

Charger circuits breaker (F5)

Outlet circuits breaker (F8)

Payload power circuits breaker (F7)

Mooring system – Electric Sys. AC Panel

External Power Feed

Air Carrier power switch

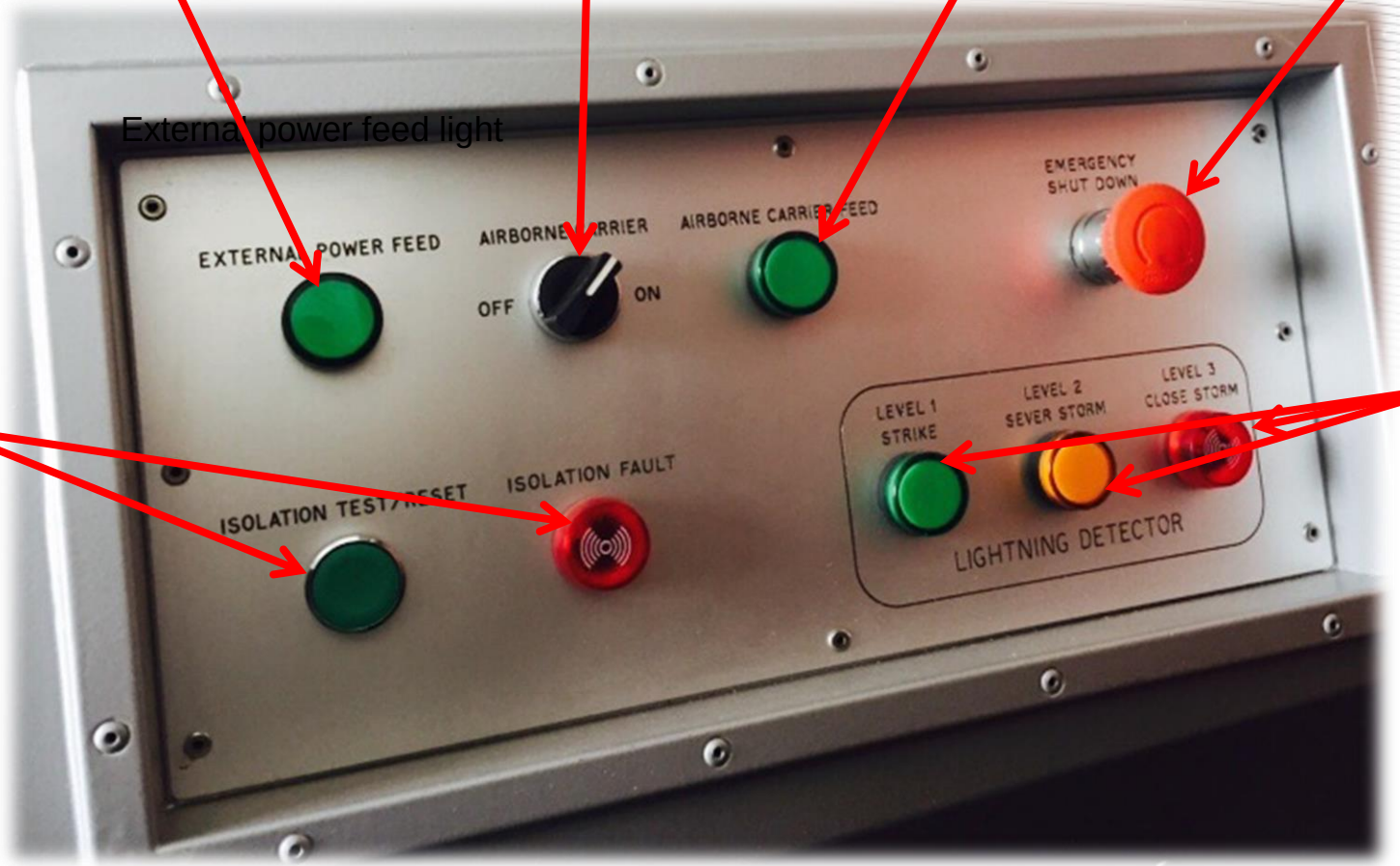
Carrier feed light

Emergency shut down

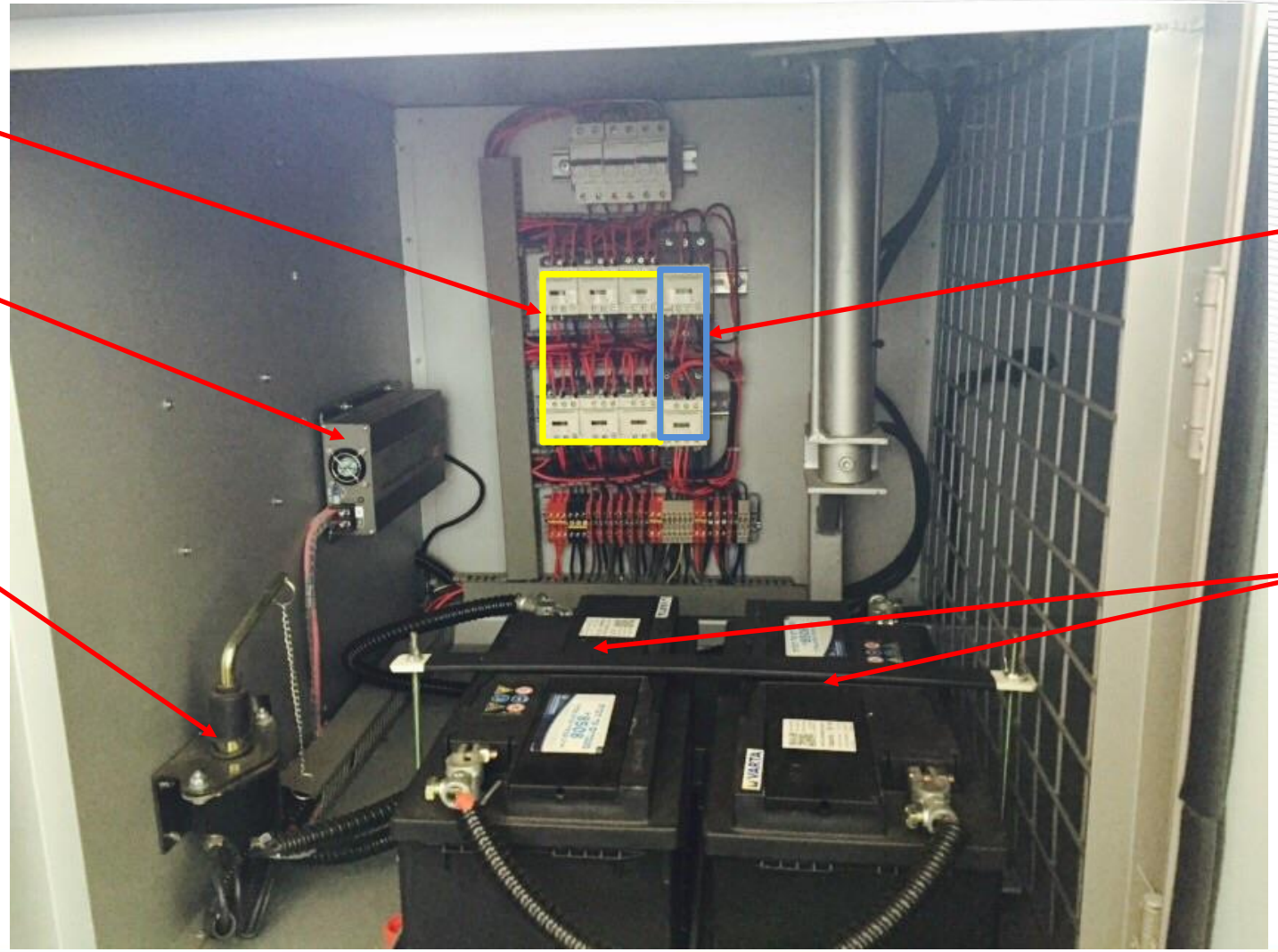
External power feed light

Isolation test/reset & alarm

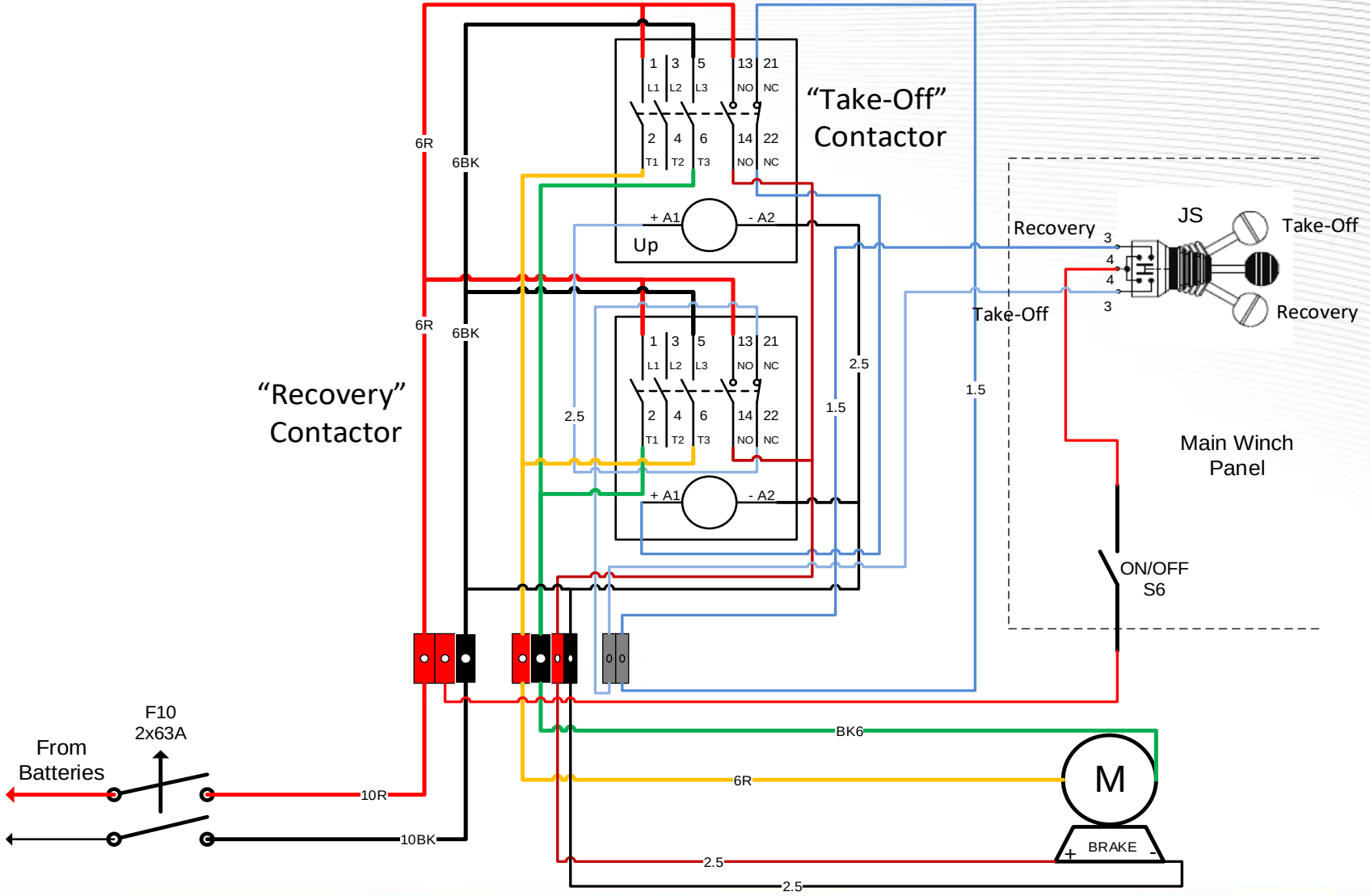
Lightning Detector Indicators



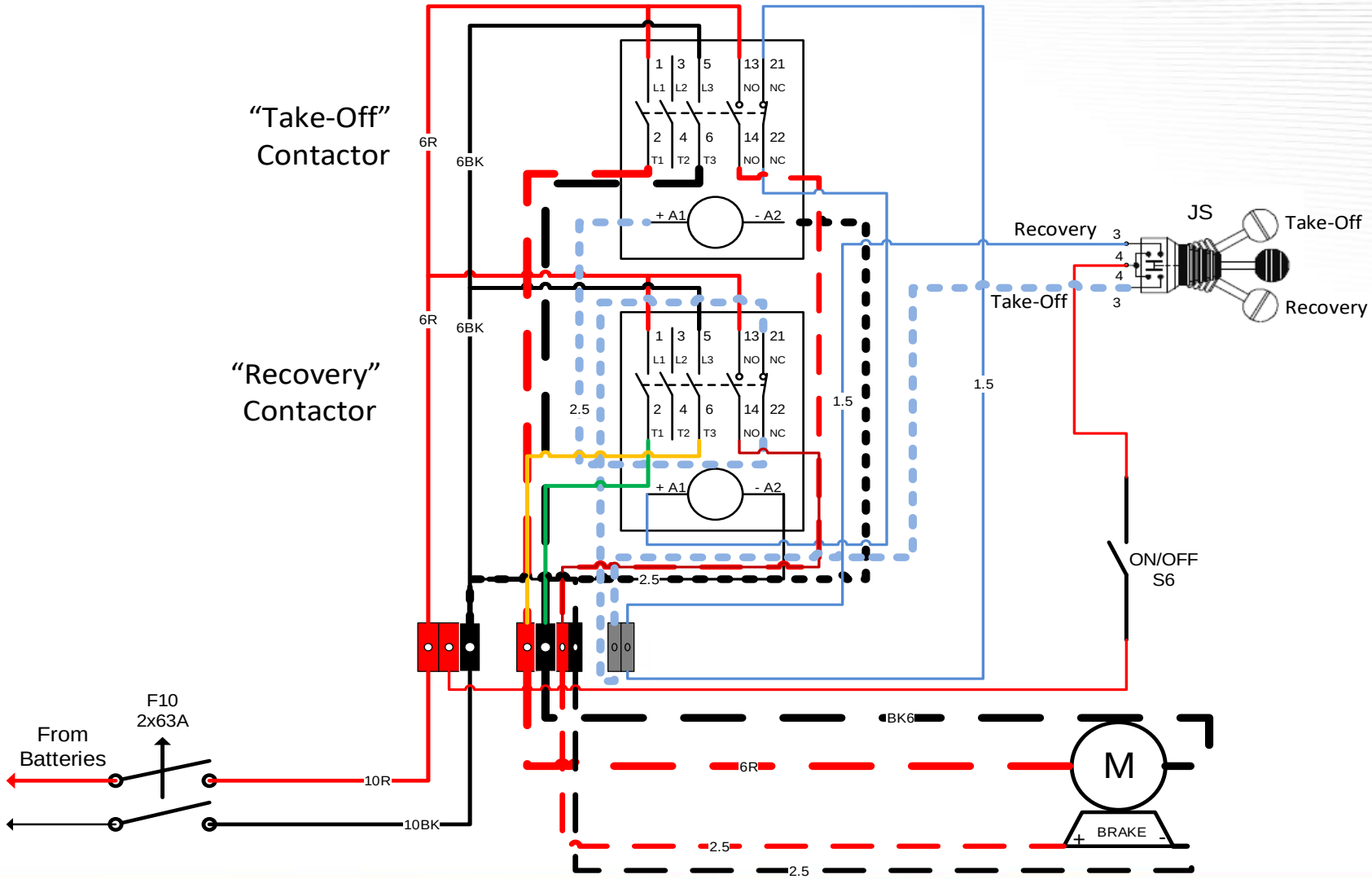
Mooring system – Electric Sys. DC Compartment



Mooring system – Electric Sys. Main Winch Command Drawing



Mooring system – Electric Sys. Main Winch “Take-Off” Command



Main winch ON/OFF switch (F6) should be set to ON and a “Take-Off” command should be applied by the Joystick.

The command goes through N.C contact in the “Recovery” contactor (To insure contrary command in parallel),

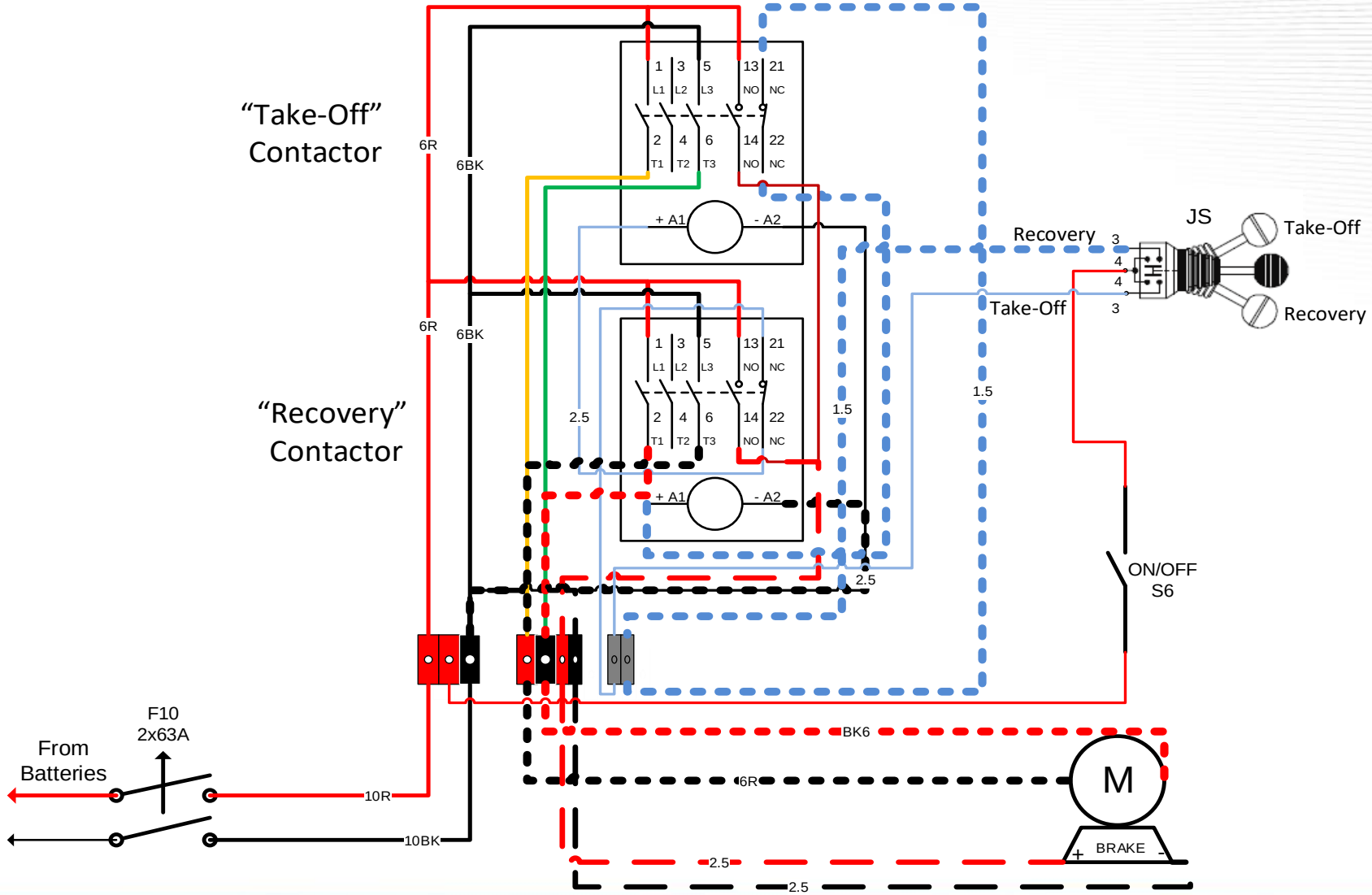
From the “Recovery” contactor the command goes to “Take-Off” contactor’s coil.

“Take-Off” Relay is switched (NC-> NO & NO->NC)

24V goes through the “Take-Off” contactor to :

1. Opens winch’s Motor Brake.
2. Turn the motor to “tack-Off” direction.

Mooring system – Electric Sys. Main Winch “Recovery” Command



Main winch ON/OFF switch (F6) should be set to ON and a “Recovery” command should be applied by the Joystick.

The command goes through N.C contact in the “Take-Off” contactor (To insure contrary command in parallel),

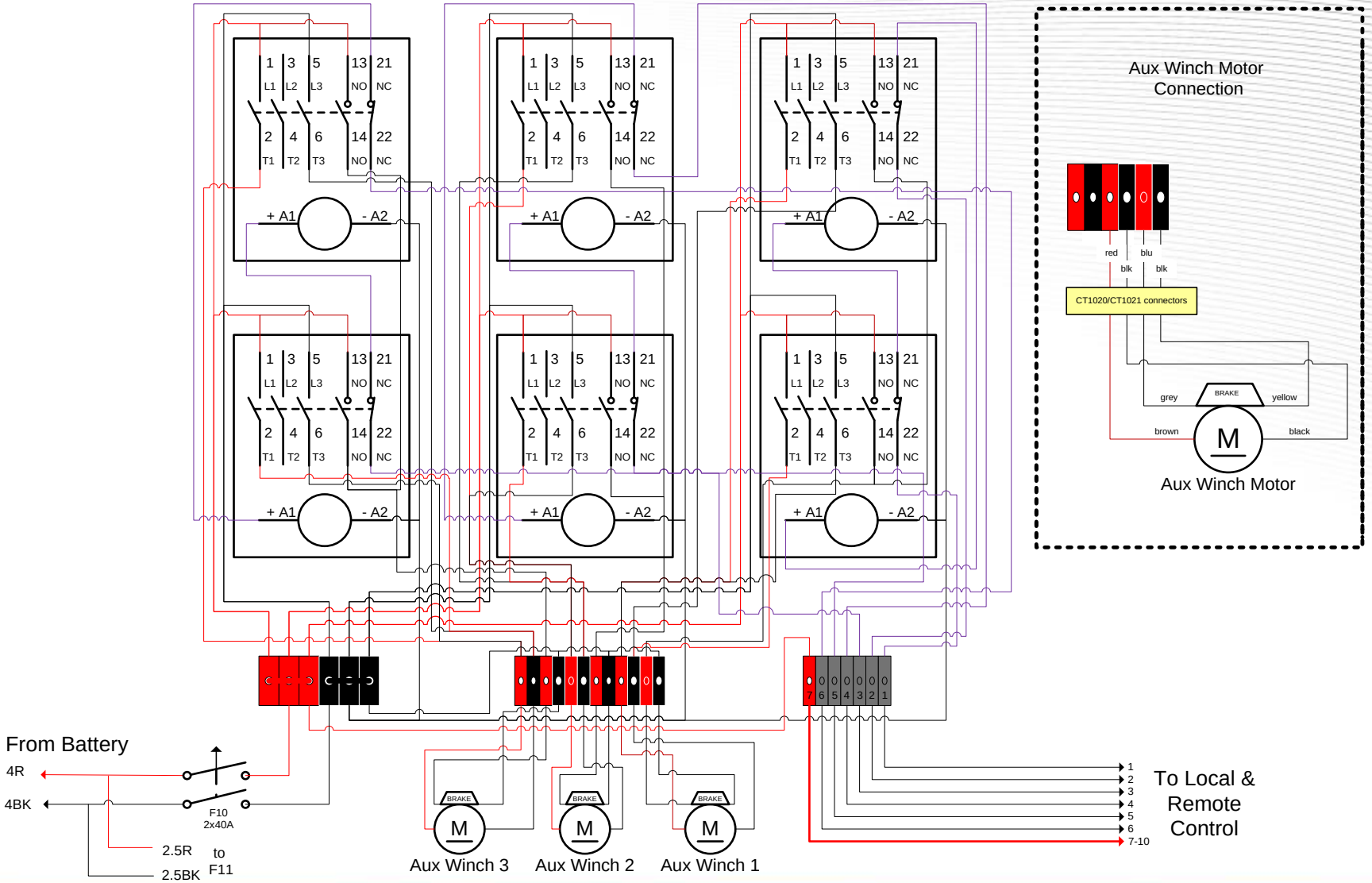
From the “Take-Off” contactor the command goes to “Recovery” contactor’s coil.

“Recovery” Relay is switched (NC-> NO & NO->NC)

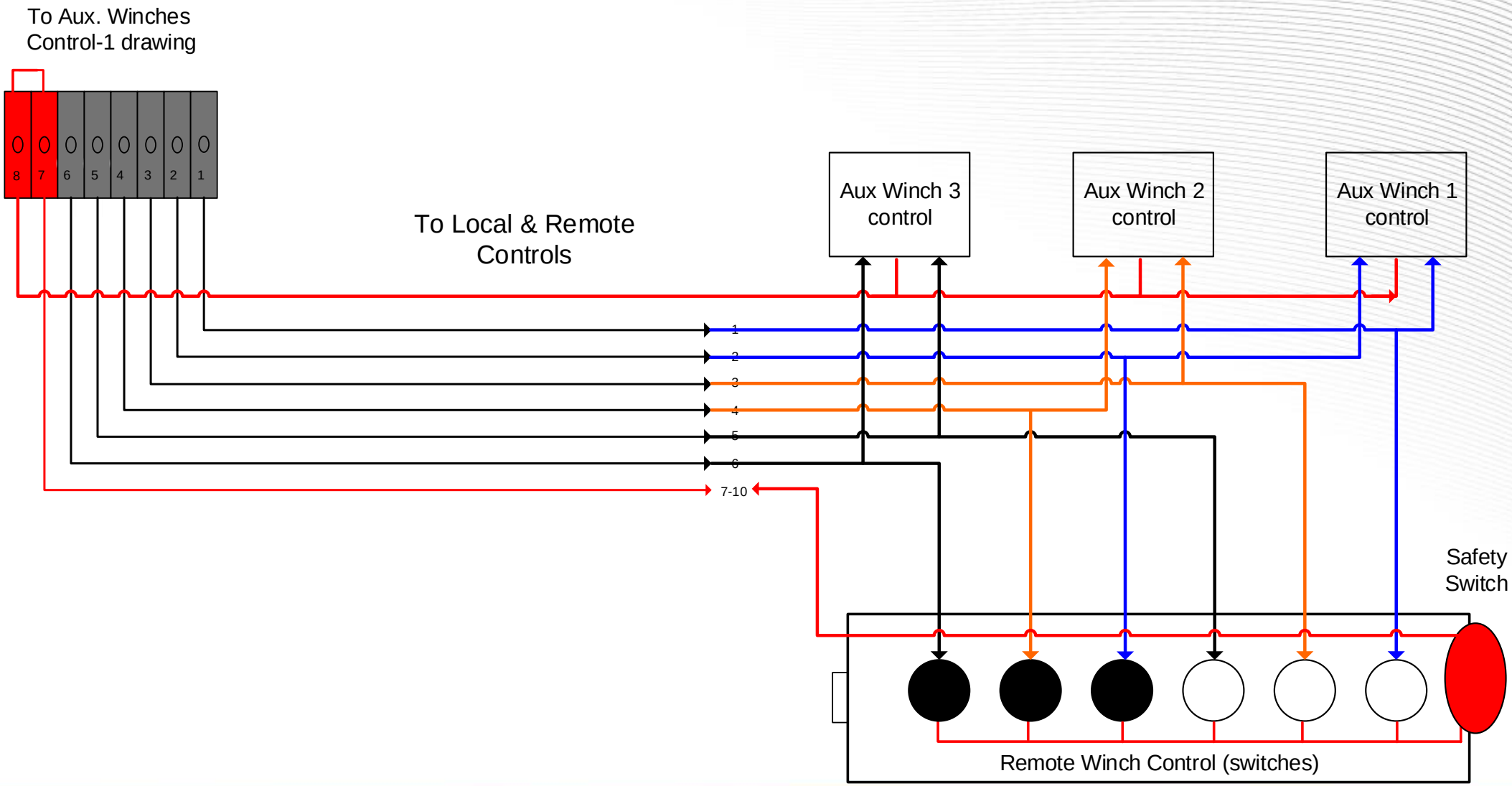
24V goes through the “Recovery” contactor to :

1. Opens winch’s Motor Brake.
2. Turn the motor to “Recovery” direction (Change polarity).

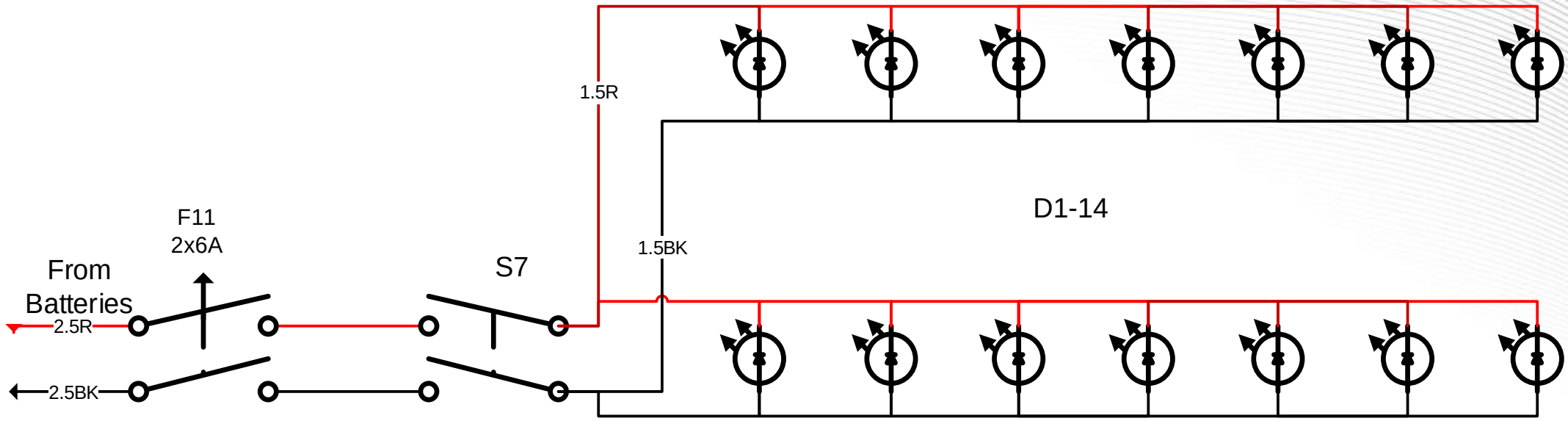
Mooring system – Electric Sys. Aux. Winches Command



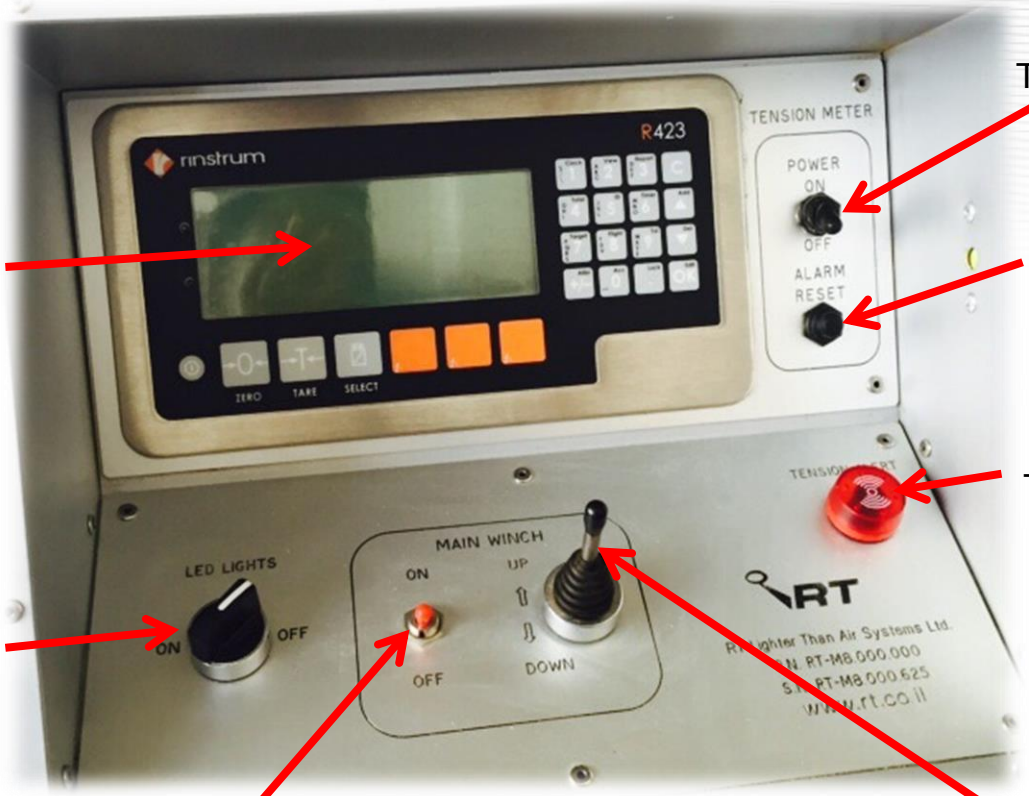
Mooring system – Electric Sys. Aux. Winches Remote Control



Mooring system – Electric Sys. Led Lights Control



Mooring system – Electric Sys. Operator Panel



Tension Meter Indicator

Tension Meter Power On

Alarm Reset

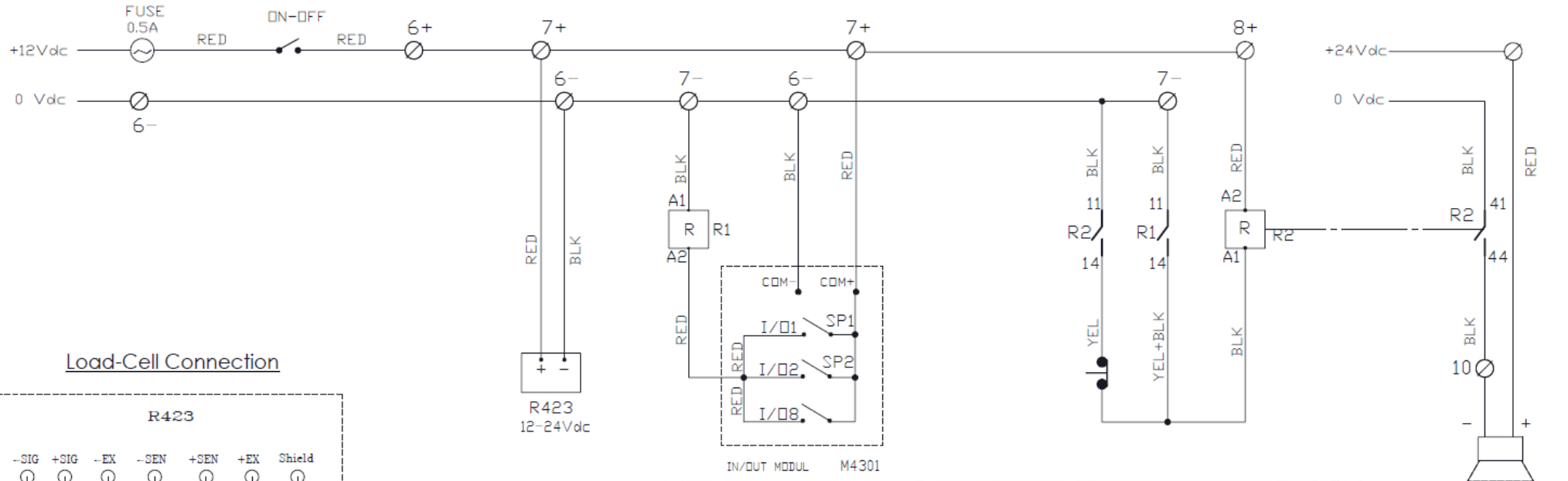
Tension Alarm Horn

Led Lights Switch

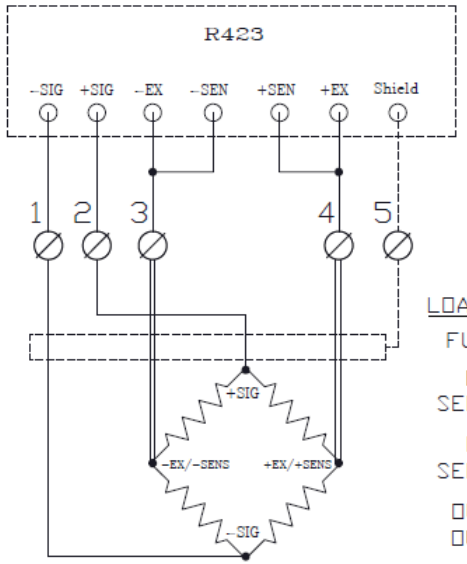
Main Winch Power On

Main winch joystick controller

Mooring system – Electric Sys. Tension Meter



Load-Cell Connection



WIRING
LOAD CELL 1330

FUNC.	COLOR
EX+	GRN
SENS+	BLU
EX−	BLK
SENS−	BRN
OUT+	RED
OUT−	WHT

Shield Transparent



Mooring system – Electric Sys. Communication E.O.P

Light Indicator and function:

1.Power light – Power is On.

2. Connection between the aerial and ground EOP – Green light Good Connection, Red light – Bad Connection, No light – No Connection.

3.Network devices connection – Green Light-there is connection through Ethernet, to near Device.



Mooring system – Electric Sys. Safety

- The main tether cable conducts 230V AC, it is strictly forbidden to work with a damaged or deformed cable.
- Before installing or servicing, always turn off all power and disconnect the main 230V AC harness from the 230V AC inlet. Electrical shock can cause personal injury or death.
- Wiring work shall be carried out only by qualified personnel. Otherwise, there is a danger of electric shock or fire.
- It is forbidden to operate the system during isolation fault, indicated by the Isolation Fault Indication and Horn.



SkyStar 180 – Technical Training

Lesson 8: PGCS - Portable Ground Control Station



PGCS - Portable Ground Control Station

The Portable Ground Control Station (PGCS) is rugged full-sized control station designed for prolonged remote control and operation of SkyStar Payloads.
The PGCS can be located near the Mooring sys. or in a distant location, connected to the Mooring Sys. via Radio wide-band data-link, or wired communications – directly to the Ethernet port at the Mooring sys.
The PGCS must be connect to the Internet for Software upgrades, TeamViewer remote control (Remote support) & Maps downloading. Can be done by physical connection using one of the RJ-45 connectors or using WIFI Hot Spot.



PGCS - Portable Ground Control Station – External Components

The PGCS built from two parts:

External section (An operator's interface):

- **ToughPad**- Includes software for communicating & controlling the payload.
- **Video Monitor** – Shows video from the payload.
- **Joystick**- Controlling the Payload's main functions.

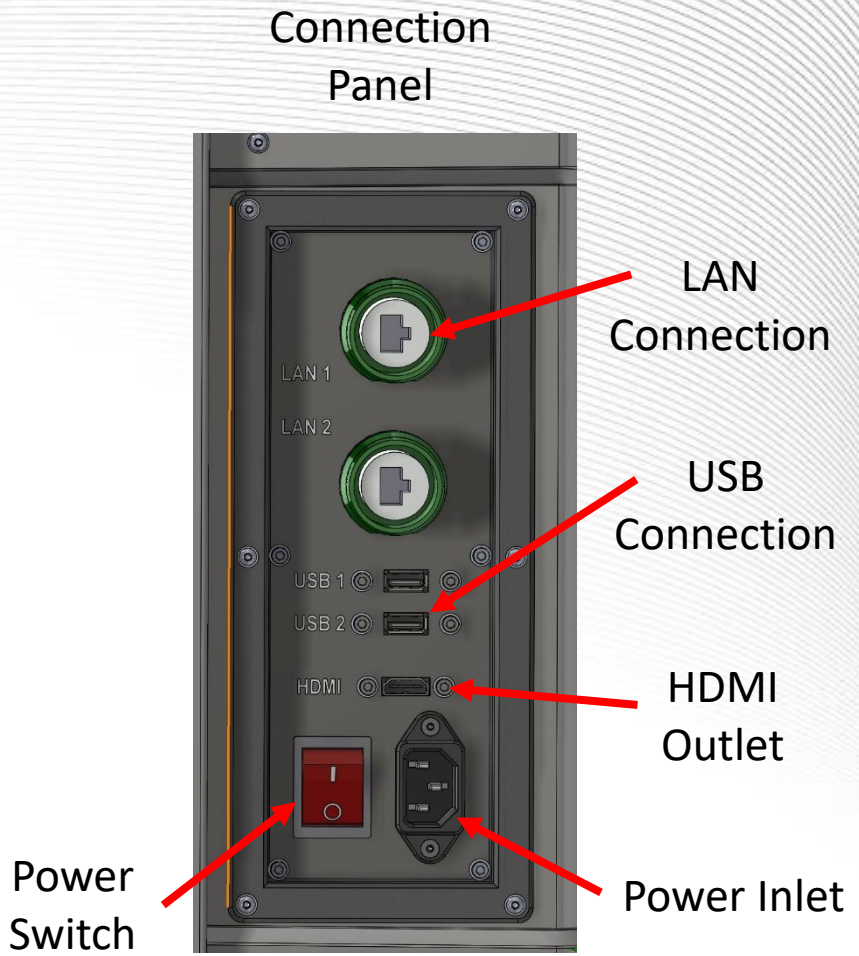


PGCS - Portable Ground Control Station - Internal Components

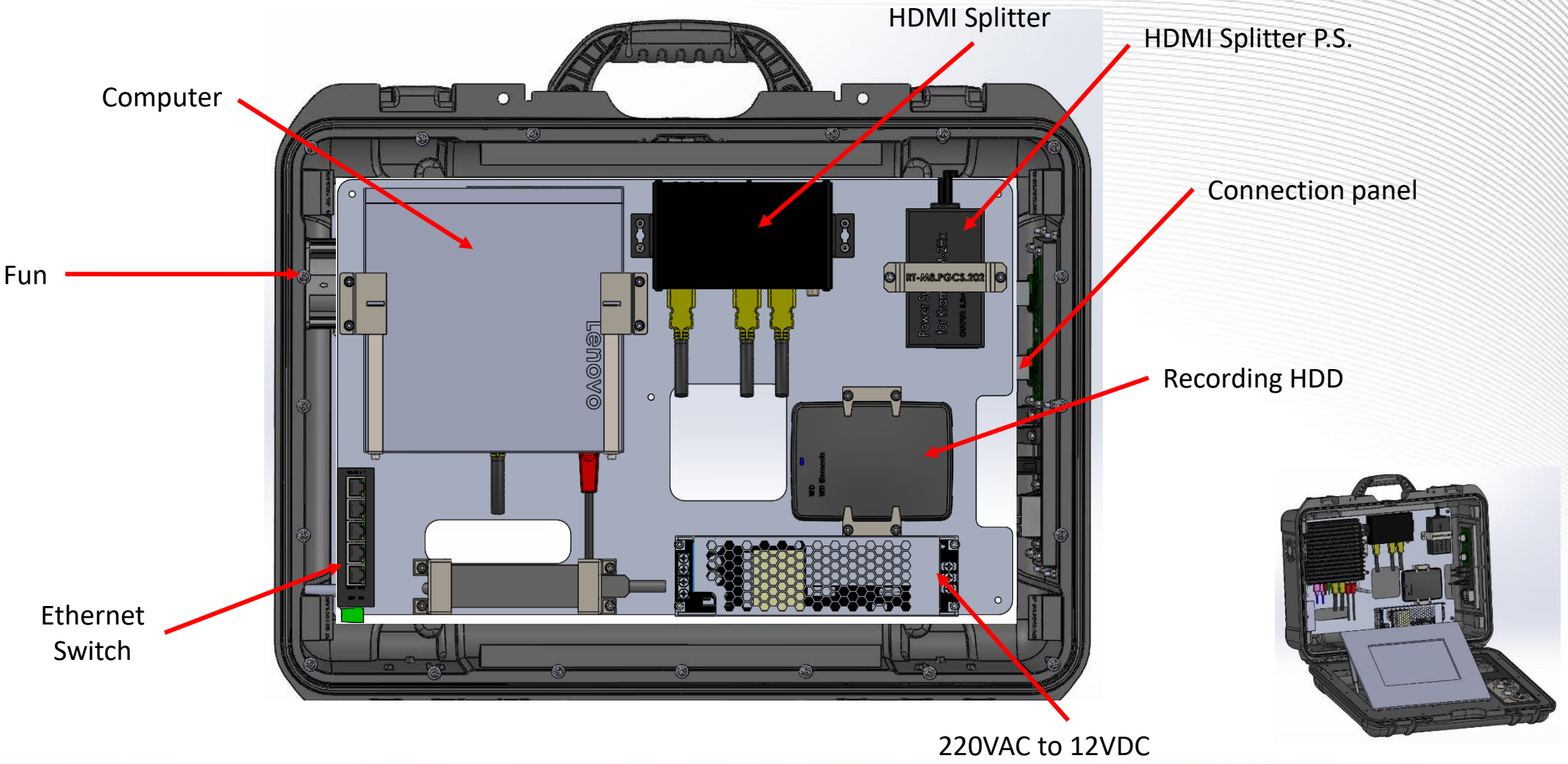
- **Internal section includes:**
  - **Power supplies -** Supply power to all electronic devices.
  - **Ethernet switch** – Enables ethernet connection between the ToughPad, Mooring Sys. And all peripheral components.
  - **HDMI splitter-** Enable to split/expand the ToughPad (digital video screen) to external & internal video monitors.
  - **Hard disk** - Enables video recording storage.
  - **USB splitter-** Enable to address the USB HD and flush disk ,use mouse to work on the ToughPad and connect joystick.
  - **Video Splitter-** Enables to split the analog video image to internal & external video monitors and DVR input.
  - **Video Decoder –** demodulate the analog video from the Ethernet line
  - **DVR –** record the analog video signal.

PGCS - Portable Ground Control Station - External Components

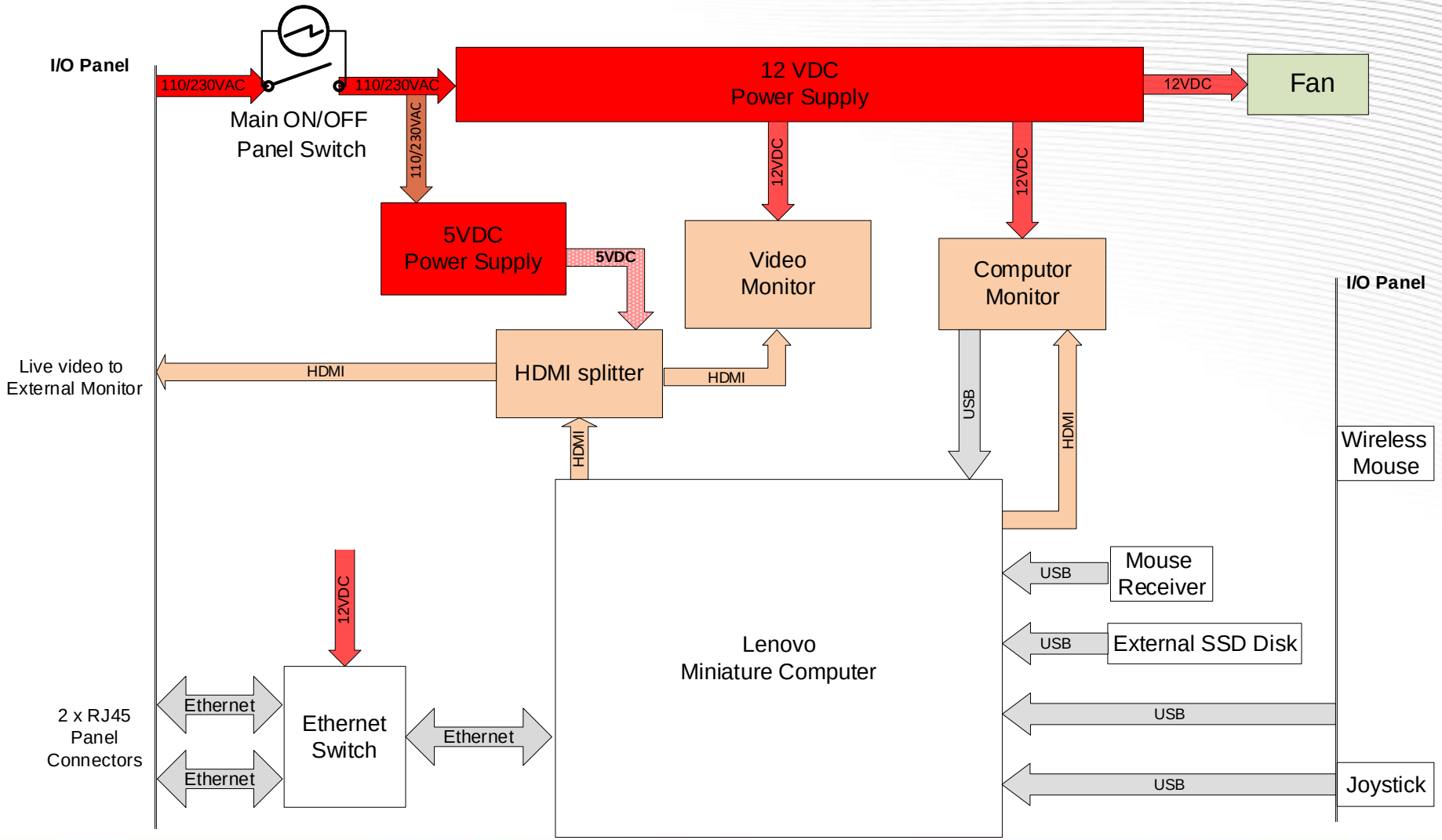




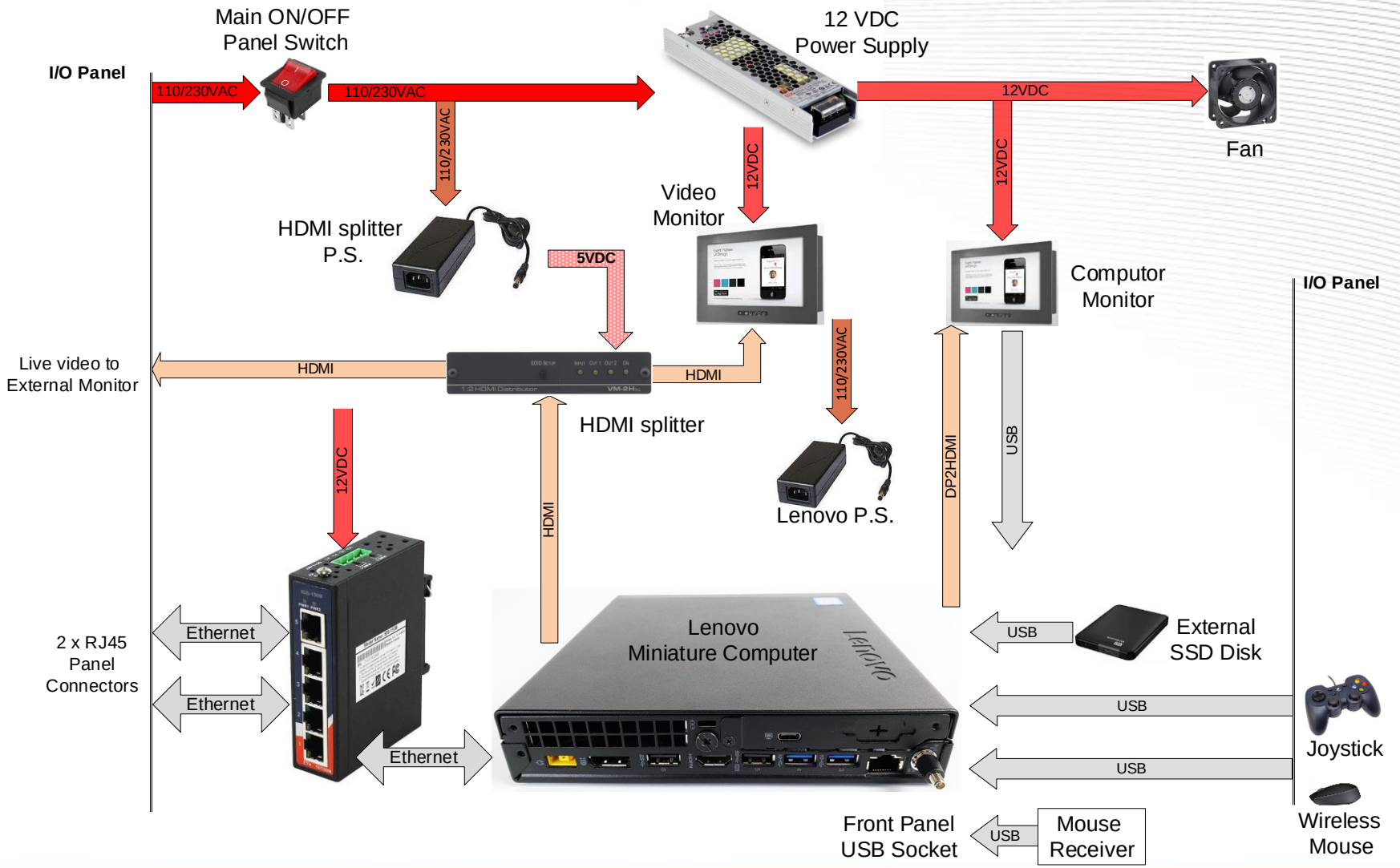
PGCS - Portable Ground Control Station - Internal Components



Portable Ground Control Station (PGCS) Block diagram



Portable Ground Control Station (PGCS) Illustration Diagram



SkyStar 180 – Technical Training

PGCS

Lesson 9: Epsilon software



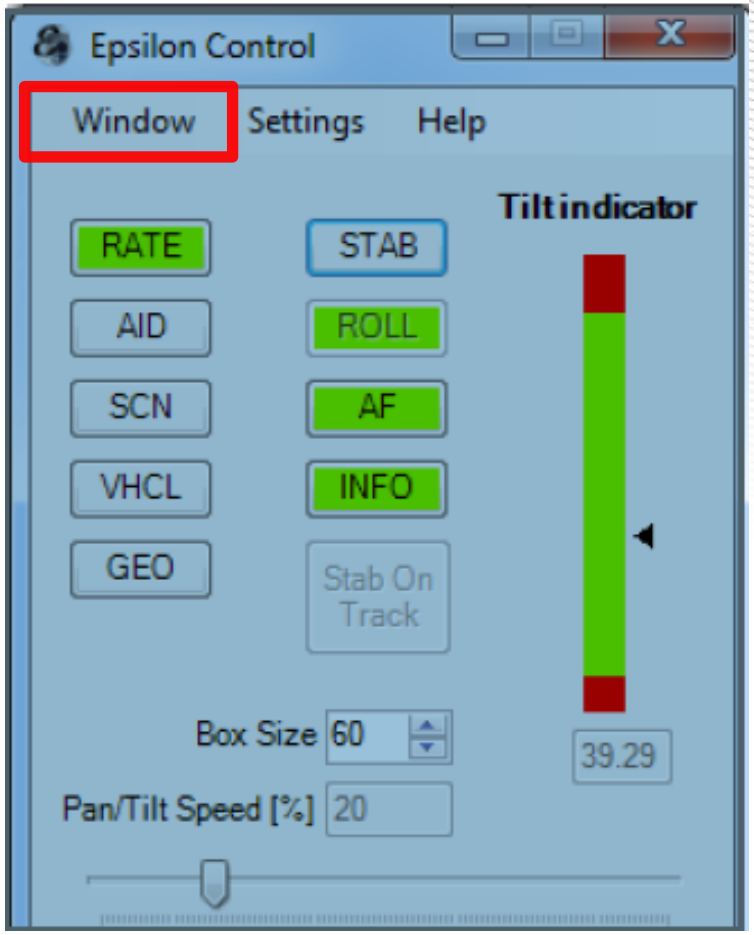
Epsilon Software – PC & Payload Connection

1. Double click on the Epsilon icon on the desktop.



Icon

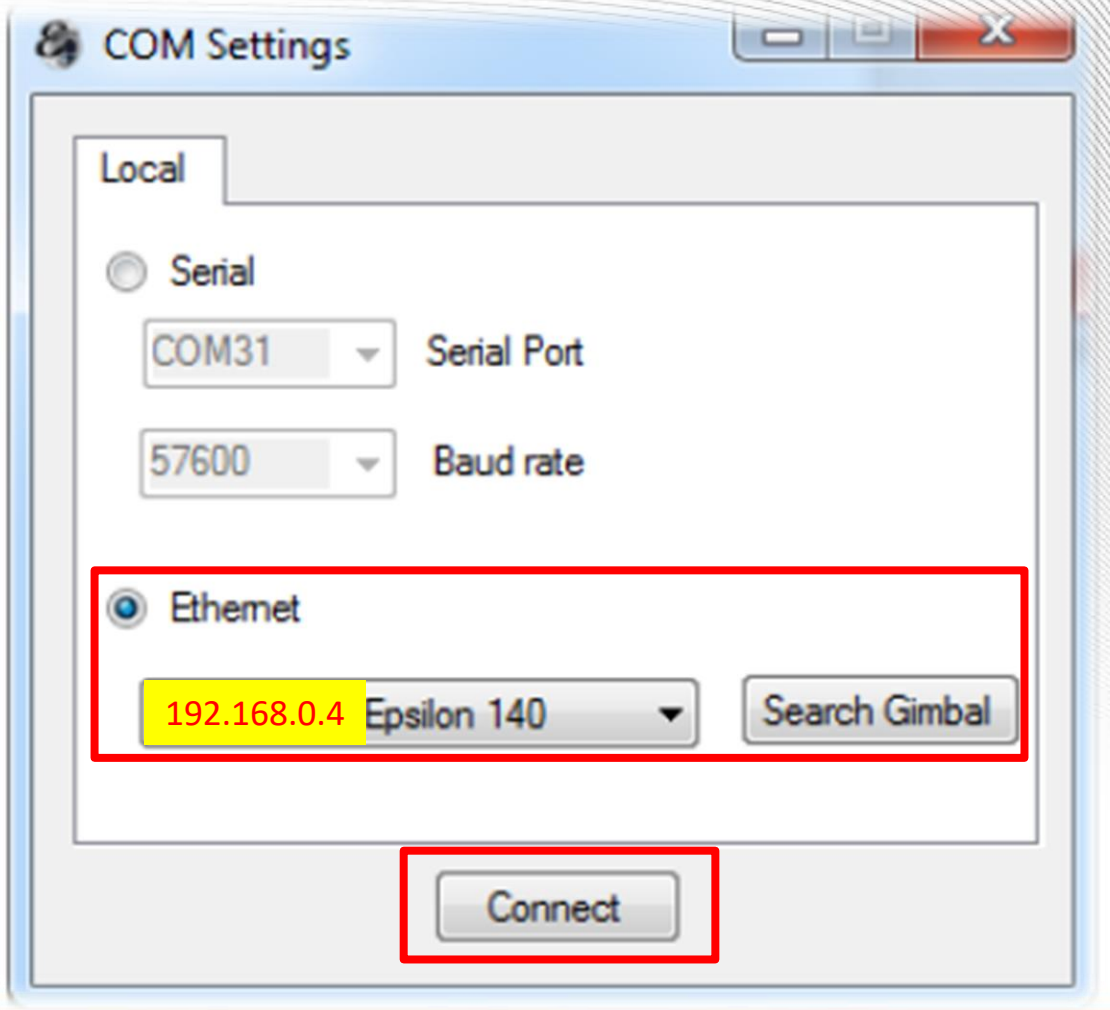
2. Connect to the payload by pressing on **Window > Connect**



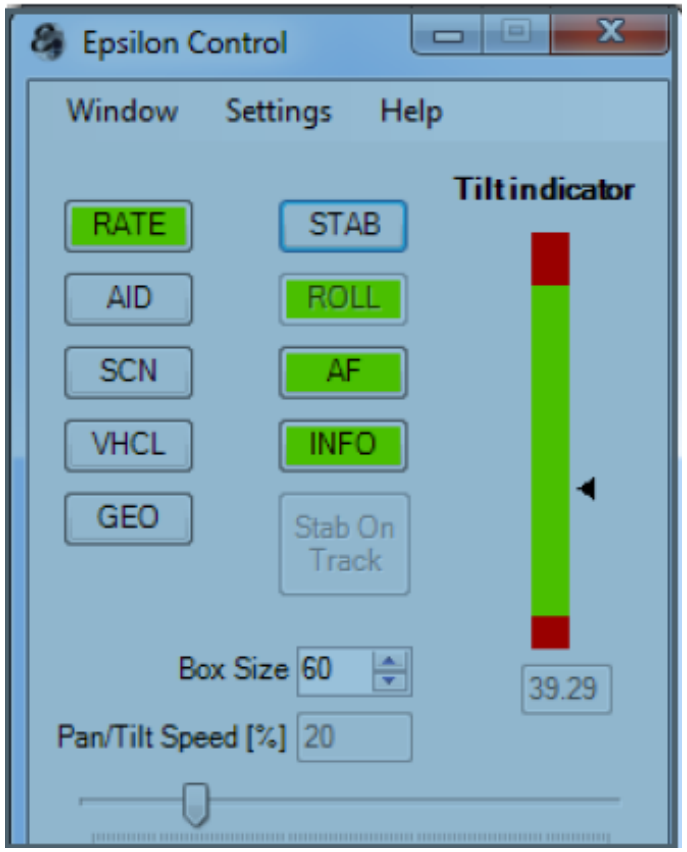
Epsilon Software – PC & Payload Connection

3. The COM Settings window will appear. Select Ethernet for communication type and press **Search Gimbal** button.
4. Click on Connect.

- After communication is established, the Status indicator on the bottom will indicate green



Epsilon Software – Main Screen Basic Functions & Buttons



RATE – Pan / Tilt manual steering mode. Uses Pan / Tilt control buttons or joystick.

AID – Pan / Tilt manual steering mode. This mode sets Pan / Tilt angular acceleration.

SCN – Scene tracking mode. This mode is used to track non-moving object.

VHCL – Vehicle tracking mode. This mode is used to track moving object such as car, boat, human and other.

STAB – Onboard image stabilization. Indicates green light when stabilization is ON.

ROLL – Onboard image rotational stabilization. The rotation limit is configured in Common settings.

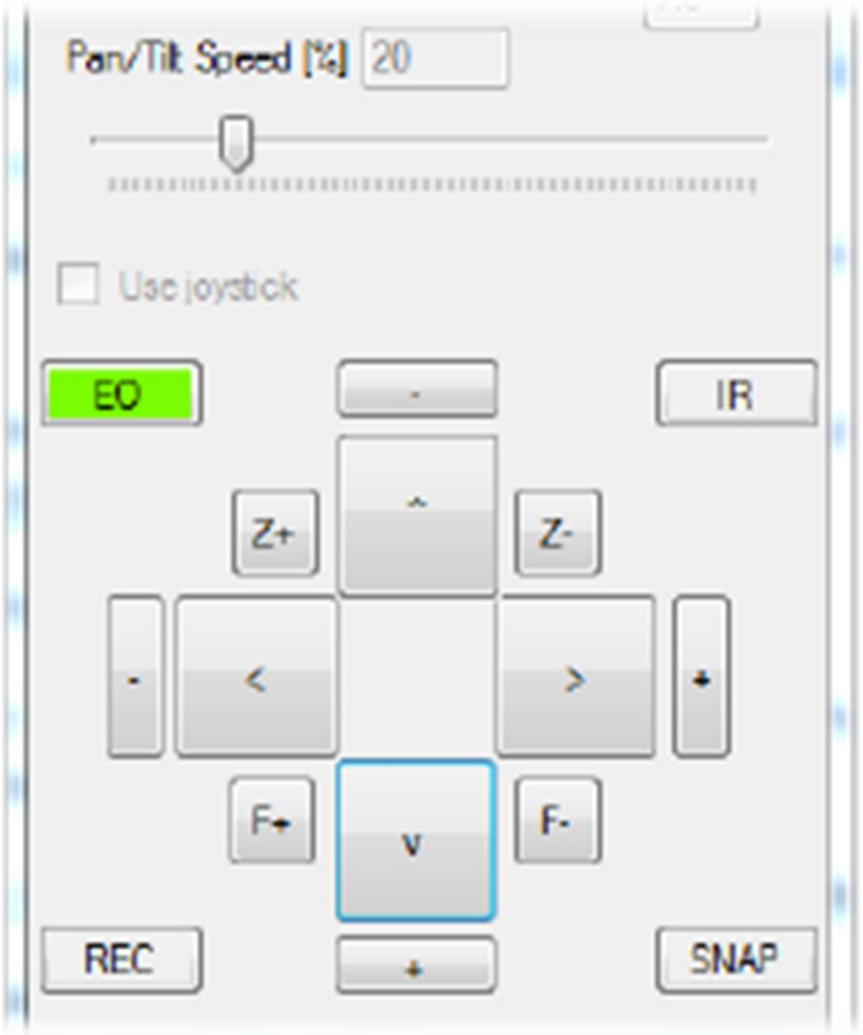
AF – Focus mode for daylight camera. Indicates green light when Focus is set to auto mode.

INFO – Onscreen information such as Tilt angle position, FOV, focus and other which is displayed on the video screen.

GEO- Enables the gimbal control by coordinates.

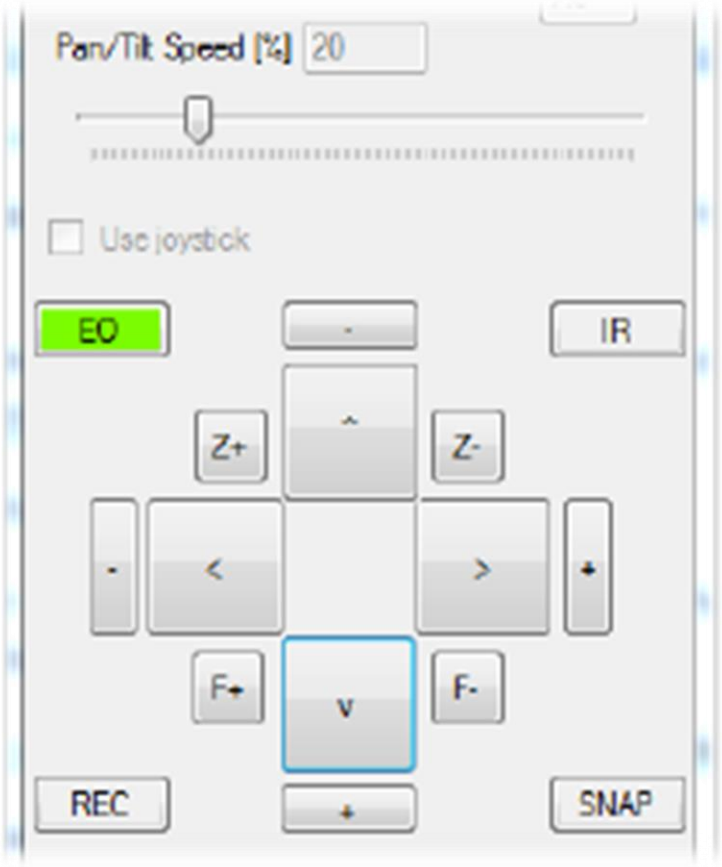
Box size – determine the size of the Tracking Box

Epsilon Software – Main Screen Basic Functions & Buttons



- **EO** – Daylight camera. Indicates green when daylight camera is selected.
- **IR** – Infrared camera. Indicates green when Infrared camera is selected.
- **REC** – enable video recording on the payload SD card (optional)
- **SNAP** – enable to take pictures stored on the payload SD card (optional)

Epsilon Software – Main Screen Basic Functions & Buttons



Trims (+ /-)

In case of small drifts on Pan or Tilt axis, the trim can be applied to the controller. Trim buttons placed near the Pan and Tilt rate buttons respectively and named as + and –

Zoom control (Z+ / Z-)

Optical zoom can be controlled by zoom control buttons.

Focus control (F+ /F-)

Focus can be controlled manually by focus control buttons

Speed limit

Pan/Tilt rate control can be limited by the limit slider

Joystick control

The Epsilon Control application supports standard USB joystick to operate the Epsilon. plug USB joystick and set Use joystick option.

Epsilon Software – Joystick Buttons





Joystick functions description MODE 1

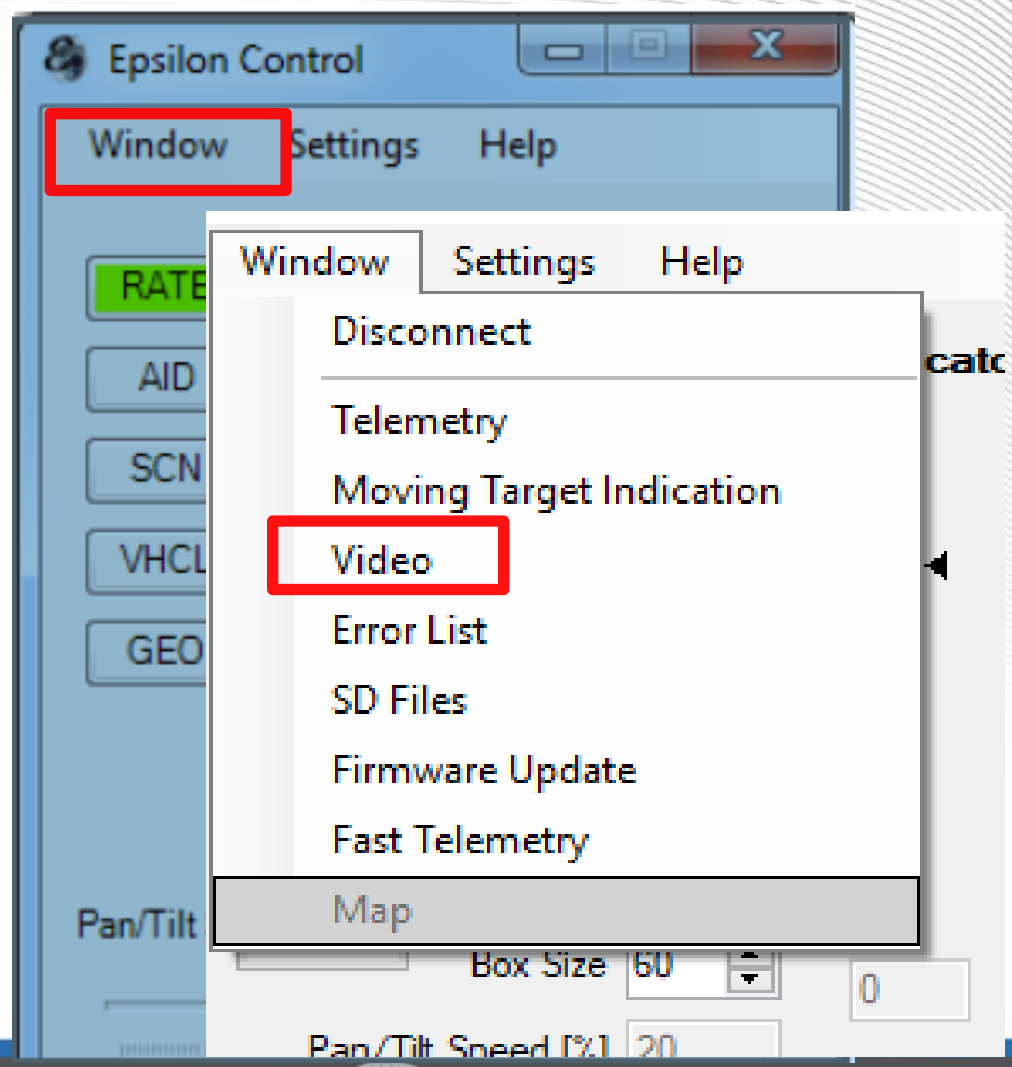




Joystick function description MODE 2

Epsilon Software – Video Display

Click on Window then click Video
The video screen will automatically prompt

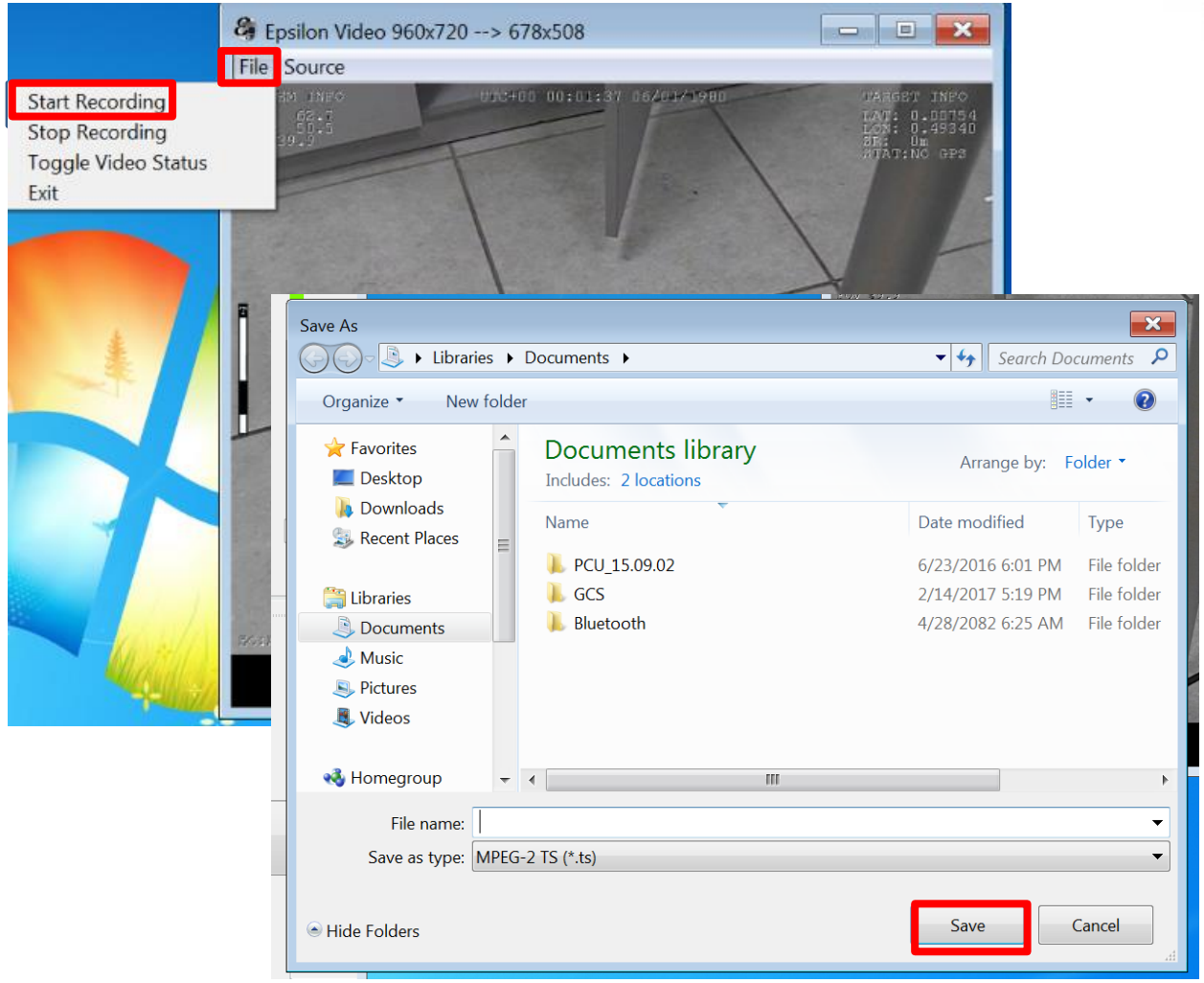


Epsilon Software – Video Screen Information

Video Screen Top side



Epsilon Software – Video Recording

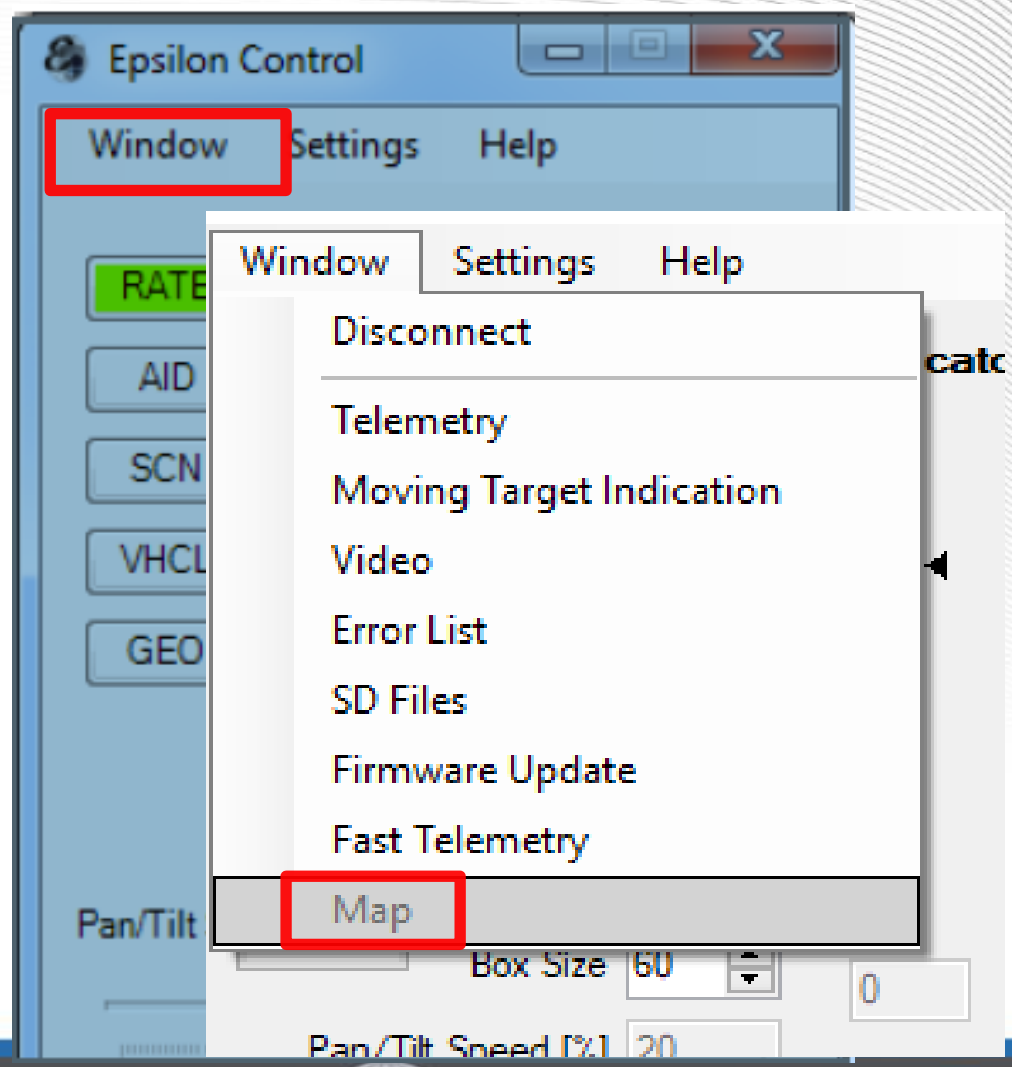


1. In the video screen click on “file”.
2. Select “Start Recording”.
3. Choose the path for the video recording file location.
4. Define the file name.
5. Click on “save”.
6. To stop select “Stop Recording”.

Recording video

Epsilon Software – Geo Location Map

To open Map view click on
Window > ***Map***
The Geo Map screen will automatically prompt



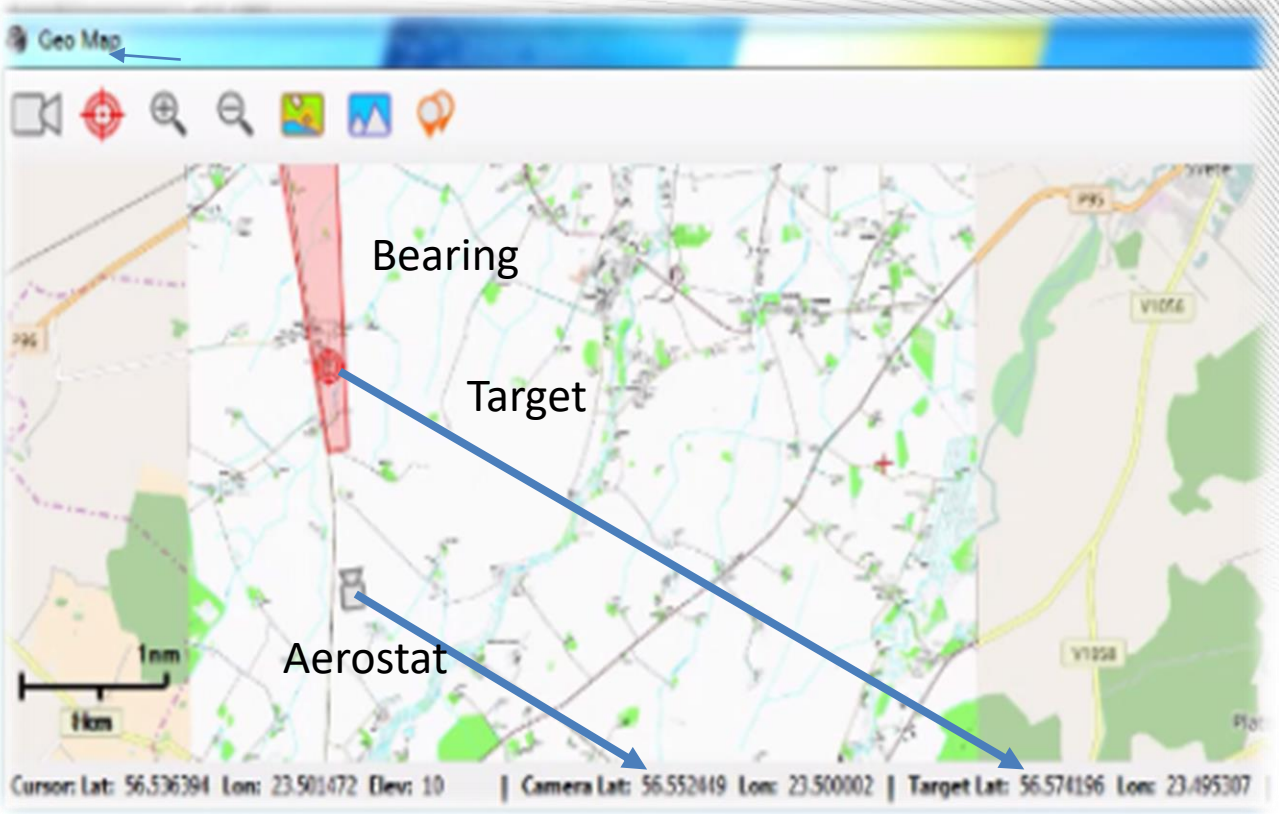
Epsilon Software – Geo Location Map

The map will be download according to the GPS information

In the bottom of the opened window operator can see:

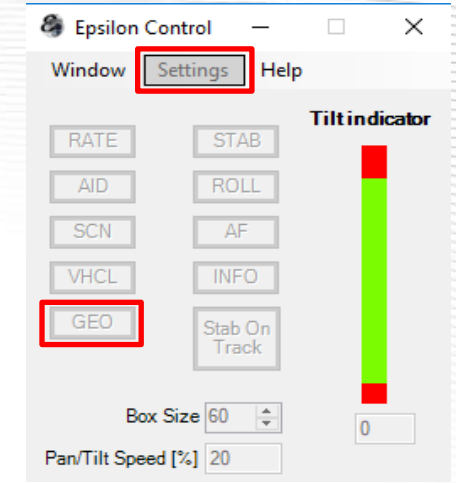
- Mouse cursor GPS location
- Camera GPS location
- Target GPS location

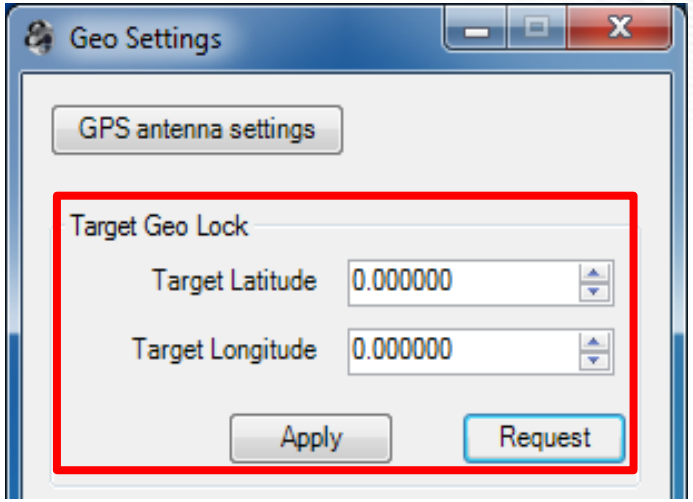
* To use online maps Ground Control station has to be connected to the internet at all times.



Epsilon Software – Map Calibration

- Lock the gimbal on the target with the known geographical coordinates (eg. cross-roads section, a house, etc.)
- Enable geo location based control of the gimbal by pressing “GEO” button in the main window of Epsilon software
- In the main Epsilon software menu select: ***Settings*** > ***Geo*** input target Latitude and Longitude in the fields under “***Target Geo Lock***” menu and press “***Apply***”

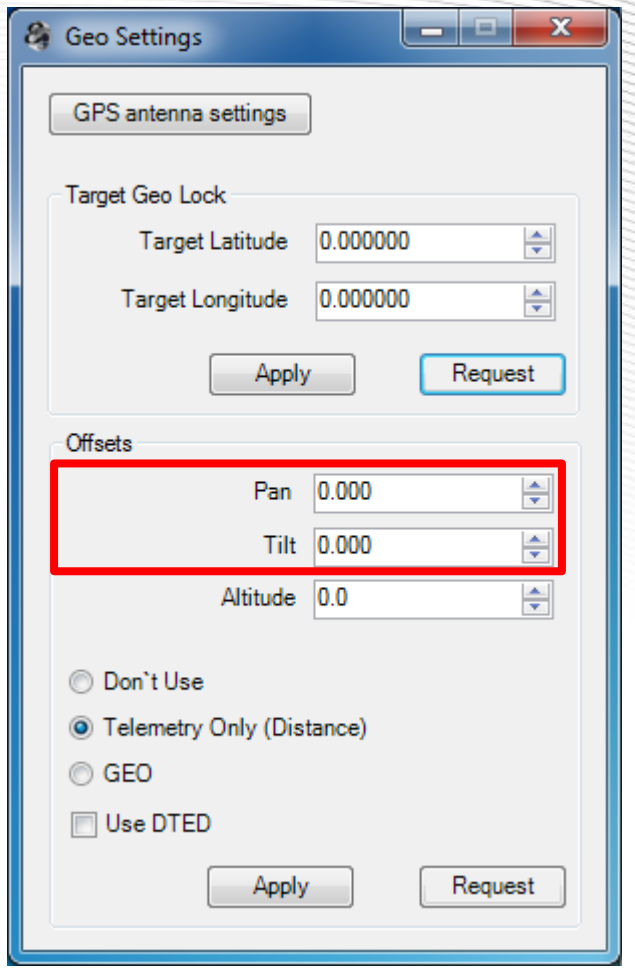




Epsilon Software – Map Calibration

- After the gimbal is locked onto the target input the “Offsets” part of the open menu by typing in “Pan” and “Tilt” offsets in degrees to center the video stream on the chosen target.

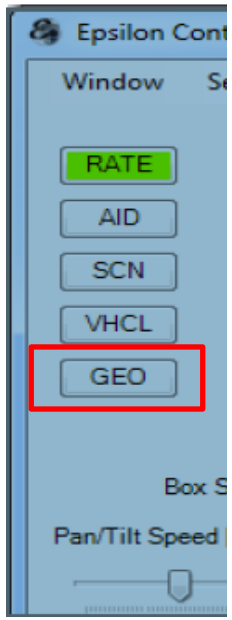
Note : the Pan/ Tilt offsets should be no more than 2 Degree

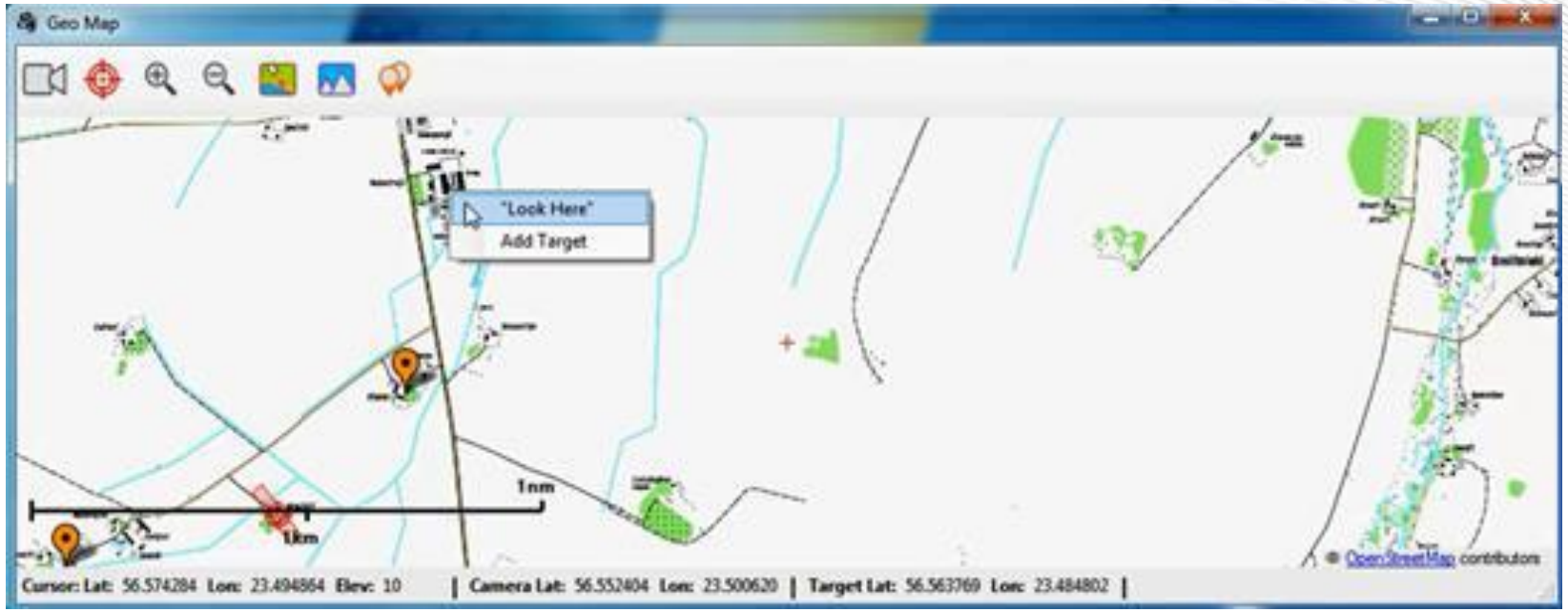


Epsilon Software – Geo Location Targets Bank

To choose a point of interest on the map operator has to left-click on the Point of interest and select “Look Here” in the opened menu. In this case gimbal will turn automatically towards the chosen target.

Note: **GEO mode must be enabled in order for this function to take action.**





Epsilon Software – Geo Location Targets Bank

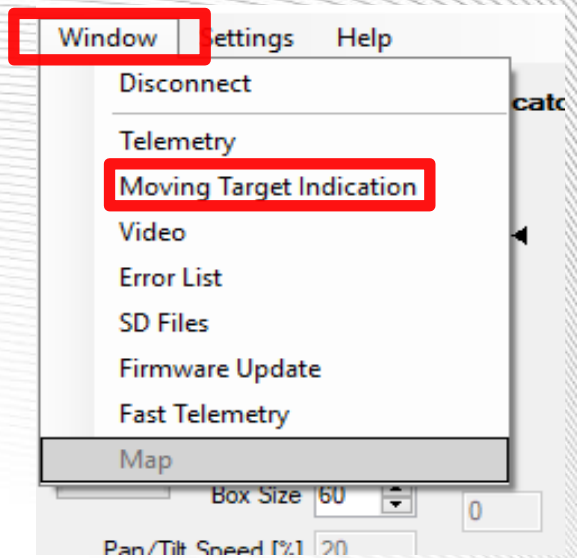
- To add targets to the targets bank from the map left-click on the target of interest and select “Add Target”.
- Target will appear on the list available by clicking “Target List” icon in the top menu of the Moving Maps Overlay Window.
- In the opened menu operator can input chosen target name, delete target, or add other target(s) by entering it’s GPS data information.

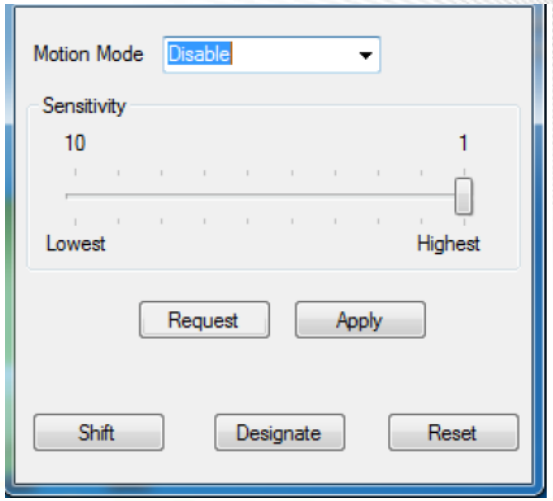


Epsilon Software – Moving Target Indicator (MTI) Setting

Moving target indicator (MTI) is optional feature, which reduces operator workload with automatic detection of moving targets. Moving Target Indicator feature has a 3 different modes:

- **Vehicle mode** – This mode works well for finding moving object that are typically 10 to 100 pixels long with relative constant speed. Not suitable for very small, very large, slow or erratically moving objects.
- **Staring** – This mode is used for small, slow, fast or varying rates objects while Epsilon is unmoved (fixed scene).
- **Aerial** – Works on a moving Epsilon for finding small, slow, fast or erratically moving objects. The mode increases video processor load to handle scene motion.

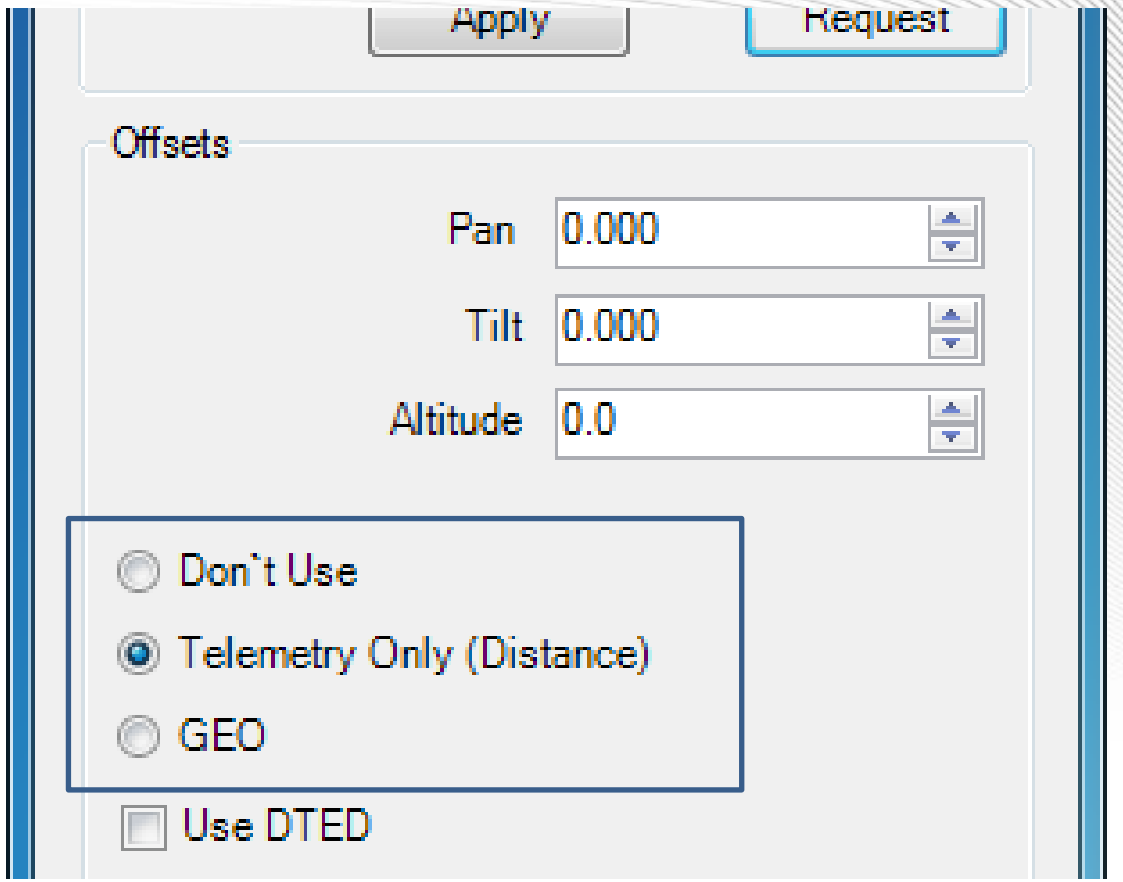




Epsilon Software – Laser Range Finder (optional)

To access Laser Range Finder Settings go to:
"***Settings***" > "***GEO***"
Laser range finder (LRF) has 3 positions:

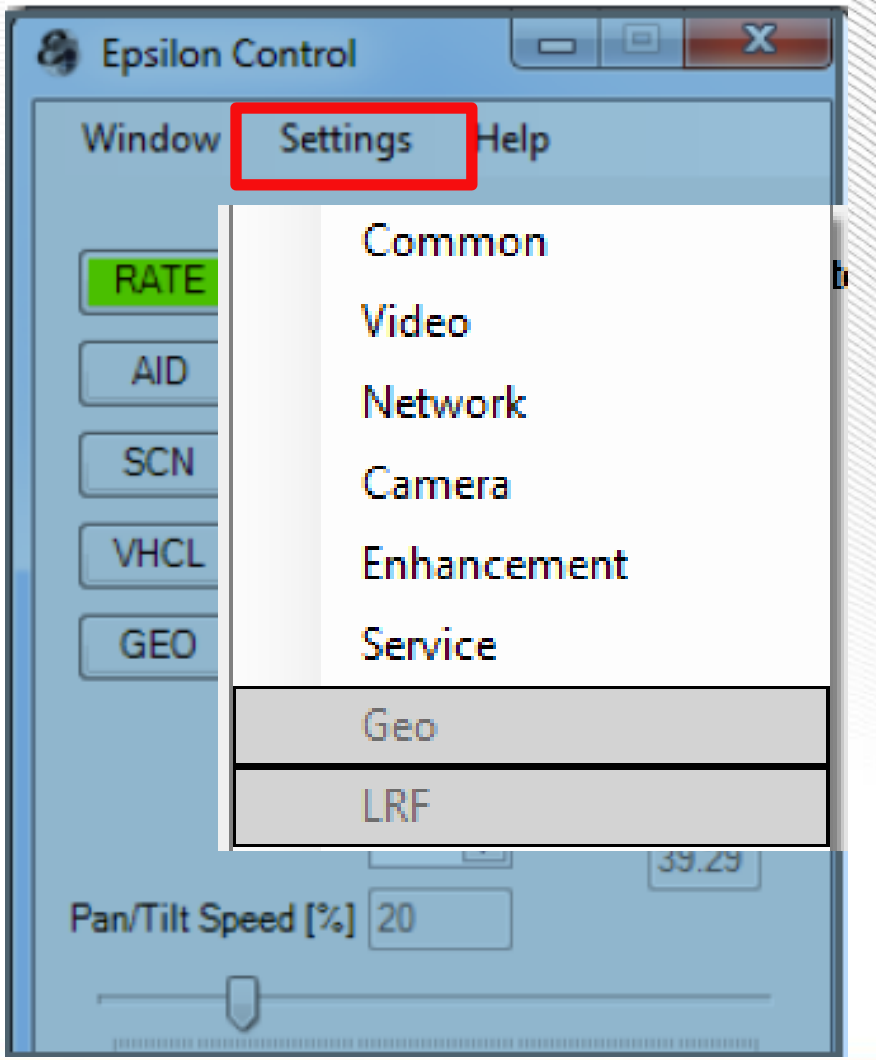
- "Don't Use" - LRF is off.
- "Telemetry Only (Distance)" – LRF is used only to measure distance and the measured value is shown in the top right corner of the video under "SR"
- "GEO" – LRF measured distance is used to determine the target location together with GPS antenna data.



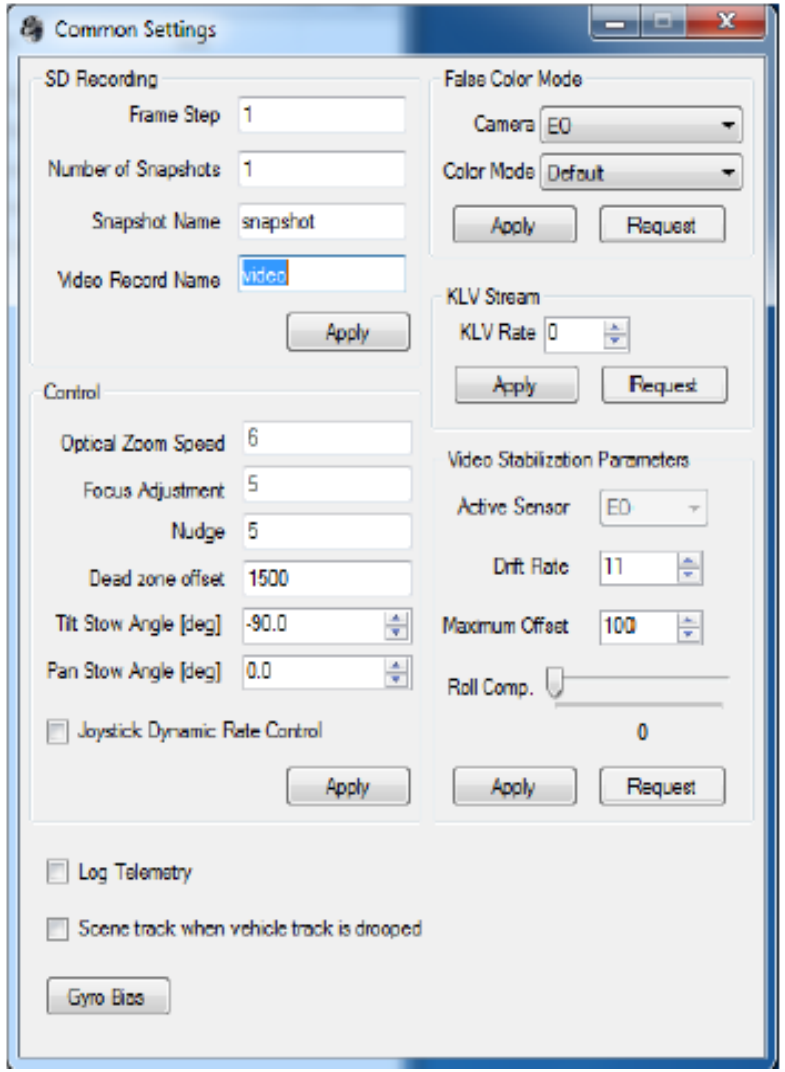
Epsilon Software – Settings Menu

Click on Settings to Open the Setting menus

Note: the following screens shows the default values for the settings



Epsilon Software – Common Settings



Control Settings

Optical Zoom Speed – Optical zoom speed limit. The speed range is from 1 to 7. Valid only for Daylight camera. IR camera has fixed continuous zoom speed.

Focus Adjustment – The focus adjustment step. The range is from 1 to 255.

Nudge – N/A.

Dead zone offset – Joystick Pan/Tilt control dead zone. Default dead zone is 1500.

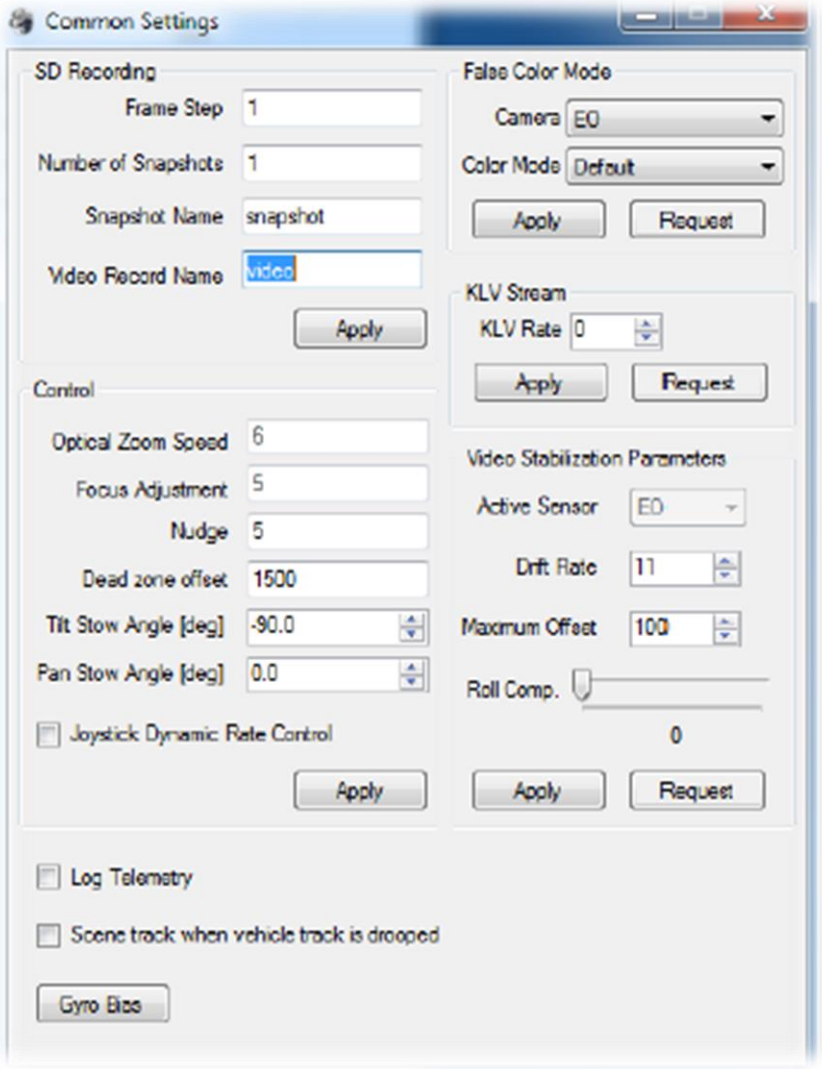
Pan/Tilt Stow Angles – currently not used. Future releases.

Joystick Dynamic Rate – set joystick Pan/Tilt control to logarithmical scale.

False color

False color can be applied to original grey IR image to make it more comprehensible. False color allows to recognize some details in the image.

Epsilon Software – Common Settings



KLV

Non relevant for Aerostat

Video Stabilization Parameters

Video Stabilization parameters can be set for each sensor. Active sensor is shown in text box.

Drift Rate – Image re-centering rate by onboard processing. Large values lead to rapid re-centering, and eliminate only highest frequencies. Small values lead to slow re-centering, keeping scene nearly still. The range is 0-255. The default value for the Epsilon is 11.

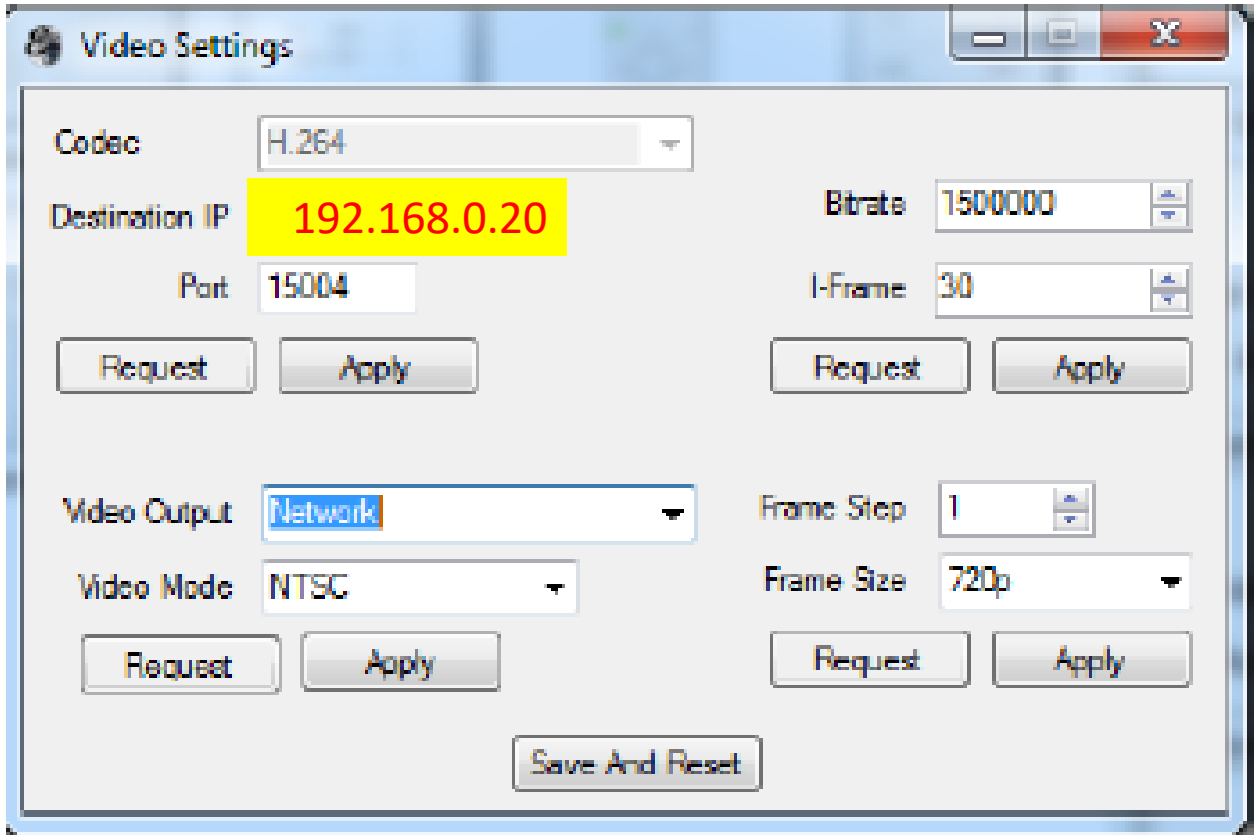
Maximum Offset – An image stabilization offset (translation limit) in pixels. The default value for EO camera is 100 and the value for IR is 30.

Roll compensation – maximum stabilization angle in degrees for cameras roll motion.

Gyro bias

Gyro Bias – Gyro drift calibration. The gimbal must not move during the calibration. This function calibrates gyro drift. Pan and Tilt motors are off for 5 seconds.

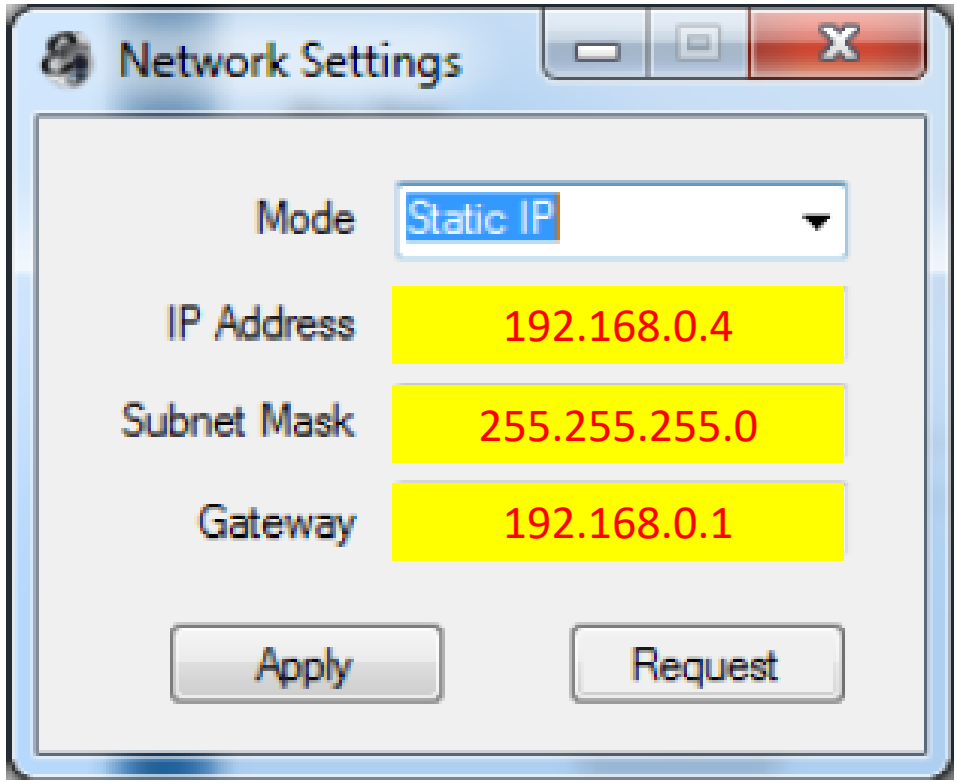
Epsilon Software – Video Settings



Codec – The default is H.264 codec for HD video.
Destination IP – The destination IP address. This address must be set the same as PC IP address which is used for digital video receive.
Port – The default port is 15004
Video Output – The default is Network for digital video. The analog or network and analog together can be selected.
Video mode –PAL or NTCS can be selected for analog video.
Bitrate – The default bitrate is set to 1500000 B/S.
I-Frame – Key frame interval. The default value is 30.
Frame Step – frame interval to be displayed. The default value is 1 which corresponds to 30 frame/s.
Frame size – Displayed frame size. The default value is 720p (960x720). SD can be selected as well.

To apply video settings press **Save And Reset** button. This operation requires reset for video processor unit. It will take around 30 seconds. The video will be lost while restarting.

Epsilon Software – Network Settings



Mode – The default mode is Static IP but DCHP can be used.

IP Address – The IP address of Onboard Image processing unit. The default IP is set to 192.168.0.4

Subnet Mask – The default mask is set to 255.255.255.0

Gateway – The default gateway is set 192.168.0.1

SkyStar 180 – Technical Training

Epsilon software

SkyStar 180 – Technical Training

Lesson 10: Payload Firmware Update



Payload Firmware Update

Firmware update allows to:

- Restore Epsilon Payload system.
- Upgrade or add Epsilon features.

NOTE: Make sure you use correct USB-COM adapter to connect Gimbal to PC. As well do not use USB hubs on your PC, Gimbal should be connected using USB to RS232 adapter straight to stand alone USB port on your PC



RS-232 to USB Adapter

Payload Firmware Update

Firmware update steps:

1. Make sure you have a firmware version that is compatible with the version of Epsilon software installed on your computer (file with a hex extension).
2. Connect the Payload directly to your computer using serial RS232.
3. Open the Epsilon software.
4. In menu bar select “Window” and choose “Firmware update” option from drop down menu.
5. In “Firmware update” window browse correct “hex” file.
6. Power on the Gimbal (Initializing process will not start).
7. Firmware update process will start and “Status” will show: “Tilt erasing” and after few seconds will show “Tilt writing”.
8. Once the software “Status” shows "Pan writing done. Firmware update is completed!" – Power off the Gimbal.
9. Payload is ready for operations.

